COMP0234: MSc Systems Engineering for the Internet of Things Project

"The Edge Negotiator

Trustworthy on-device small-language-model control of urban traffic signals, with a provable accident audit."

by

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Abstract

When a signalised junction is involved in a collision, someone must be able to prove what the controller knew and what it decided. Language-model traffic controllers are now competitive in simulation, but every published system is served from the cloud and none can answer that question. This dissertation asks whether a language model small enough to live in the signal cabinet can control real London junctions competently *and* bind every actuation into a record an investigator can trust. The second question is what, if anything, fine-tuning contributes.

I build the full stack on one consumer device: a small language model served by Microsoft Foundry Local within a 6 GB envelope, proposing signal phases behind a MaxPressure shield and anti-starvation floor, with every decision Ed25519-signed into a per-junction hash chain and reconstructable after a collision into a cited UK-law evidence pack. I distil a frontier teacher into 0.6B and 3.8B students with QLoRA, compile to int4 ONNX, and evaluate offline against a hash-frozen holdout and closed-loop in SUMO on measured London topologies over thirty training-disjoint seeds per cell.

Several results stand out. Distillation *saturates at 0.6B*: the 3.8B student is statistically indistinguishable from the 0.6B one offline and closed-loop, and four of the five catastrophic fine-tuned cells fall among the trajectories they share. Closed-loop delay *decouples* from decision accuracy on networks that cannot clear, where a 42-point accuracy gain yields no significant delay improvement inside the shield's envelope. A three-seed finding that MaxPressure is worse than a naive fixed-time floor on *every* real London topology fails to replicate at thirty paired seeds, its sign varying with the network without my isolating which property carries it, so I report that retraction rather than the headline it would have made. Fine-tuning's value emerges only when the test network can flow. On a Bloomsbury grid left at its default demand, the distilled and stock controllers differ by less than two points because under 15% of vehicles clear. On the same network calibrated so that 87% clear, the distilled controller *beats* the fixed-time floor by 8.8% while the stock one loses to it by 11.9%, which is an *18.3% head-to-head improvement that holds on 30 of 30 seeds* (p=2.0×10^{-21}), and against the stronger MaxPressure comparator the same distilled controller draws level to within 0.03%. Applying the identical calibration to the corridor where the controller looks worst does *not* reproduce that effect. The head-to-head there is −3.7% and not significant, so the large effect is a single-network result, and I report the failed replication alongside it. Network gridlock had been masking the effect, which suggests published effect sizes in this field may be compressed by the same cause. Fine-tuning also raises *authorship*: the model's share of logged decision ticks rises from 0.39–0.62 to 0.56–0.77, while its divergence from MaxPressure among the decisions it authors falls from 0.46–0.60 to 0.04–0.17. That matters for accountability: an audit of the stock deployment could attribute two decisions in five to the learned component, against four in seven after distillation, a deterministic timer authoring the rest. I also state the deflationary reading this invites, namely that distillation partly taught the student to imitate the classical reference, and bound it with the available evidence. Seven collision scenarios reconstruct end-to-end with tampering caught, an adversarial trust battery, scored by offline replay with gating off, catches most blatant misreporting and is evaded by a calibrated stealth attacker, and the nulls, latency exclusions, and specialisation costs are reported with their mechanisms rather than omitted.

Acknowledgements

I thank Dr Akin Delibasi for supervision that consistently pushed this work towards harder evidence. The throughput-alongside-delay test, the multi-seed robustness requirement, and the scope expansion into fine-tuning all originate with him. I thank Lee Stott (Microsoft) for the policy-driven architecture guidance that shaped the authorisation layer, and specifically for the objection that became §6.11: that a model cannot be judged on a task whose rules were never put in front of it. Testing that objection properly changed what this dissertation is able to claim. I am grateful to UCL Computer Science for access to the departmental GPU cluster, on which the 3.8B student and its five ablation variants were trained.

Contents

Terms used in this dissertation

The first three are measured quantities this dissertation turns on, and they have different denominators; confusing them is the easiest way to misread the results. The one-line forms below are for reference while reading. Their formulae, denominators and caveats are given where they are derived, in §3.3.

Authored share. Of logged decision ticks, the fraction whose served phase originated with the model. The accountability number.

Proposal acceptance. Of ticks where the model returned a well-formed proposal, the fraction served. Not a measure of authorship: the model is not consulted on every tick.

Divergence rate. Of the decisions the model authored, the fraction that differ from MaxPressure's choice.

Congestion gate. The rule deciding whether the model is consulted at all. Below a threshold of halting demand, MaxPressure serves the tick directly.

Anti-starvation floor. A best-effort minimum-service rule that force-serves any approach passed over beyond a bounded window, overriding both the model and MaxPressure. It is suspended during an admissible emergency preemption, and records a violation when two phases reach the threshold together. It authors a substantial minority of logged decision ticks.

Shield. The safety layer: phase masking, MaxPressure substitution for invalid output, and the floor. A sound heuristic composite, not correct-by-construction in the formal sense (§2.5).

Substitution path. The part of the shield replacing an invalid or absent proposal. Measured at 0.0% of decisions almost everywhere, which is what stops "the model matches MaxPressure" from being circular.

MaxPressure. The classical adaptive baseline. It serves the phase of greatest *pressure*, upstream queue minus the downstream queue it would discharge into, so it declines to serve a long queue whose exit is blocked.

Fixed-time floor. The naive comparator: each network's own default cyclic program, with no adaptive controller. The standard every deployed junction already meets.

Longest-queue-first. A third classical baseline serving the largest upstream queue with no downstream term. A different rule from MaxPressure, though measured indistinguishable from it here.

Teleport. SUMO's gridlock-escape mechanism, which removes a vehicle stuck beyond a timeout. A high count means the network is failing rather than flowing.

Demand calibration. Subsampling demand until the network clears, so controller differences are measurable rather than compressed by gridlock.

QLoRA. The fine-tuning method: low-rank adapters trained against a 4-bit quantised base, which makes training feasible on the GPU that also serves the model.

int4 ONNX. The served weight format, compiled to fit the 6 GB envelope.

Foundry Local. Microsoft's on-device serving runtime, and a fixed constraint of this project rather than a choice.

Ed25519 hash chain. The audit structure. Each decision record is signed and linked to its predecessor by a hash, so later alteration breaks the chain detectably.

LOTO. Leave-one-topology-out: a control student trained with one network's states removed, testing whether gains are topology memorisation.

Stock, ft. *Stock* is the same base model through the identical compilation pipeline but not fine-tuned; *ft* is the fine-tuned student. Comparisons are always against the identically-compiled control, never the vendor artifact.

Chapter 1

Introduction

1.1 Motivation

Traffic signals are the most consequential actuators most cities own, and the research frontier has begun handing them to large language models: GPT-4-distilled controllers, RL-tuned reasoning models, cooperative agents [14, 33, 36]. Every one is evaluated against a cloud endpoint, yet the junction is an edge environment: no guaranteed backhaul, a hard memory envelope, and a statutory duty, rare among LLM application domains, to account for itself after a collision. This dissertation takes the four research questions (§1.2) together: an accountability record concerns the certified component only so far as that component authored the behaviour.

The work is conducted with and for Microsoft, which fixes the serving stack: Foundry Local on a 6 GB consumer GPU, capping the model ladder at roughly 4B parameters. That constraint is the research object: what *this* deployable envelope can do, not what a datacentre could.

1.2 Research questions

RQ1 Can an on-device SLM (≤4 GB, Foundry Local, RTX 2060-class), behind a deterministic safety shield, match or beat classical baselines (fixed-time, MaxPressure) on measured London topologies in SUMO?

RQ2 What does distilling a frontier teacher into the SLM buy offline and closed-loop, and at what model scale does the benefit saturate?

RQ3 Can every actuation be bound into a tamper-evident audit that, after a simulated collision, yields a UK-law-grounded evidence pack rather than an unaccountable log?

RQ4 Does a trust-coefficient mechanism over neighbour reports withstand honest, blatant, and stealth misreporting?

1.3 Contributions

1. **The first documented end-to-end chain** from QLoRA fine-tuning through int4 ONNX compilation to Foundry Local serving of a custom model, with equal aggregate holdout accuracy across CPU and GPU execution providers, and three serving-reliability findings with mitigations (§4.4, §6.12).

2. **A scale-saturation result, at two levels**: frontier distillation lifts a 0.6B student to the teacher-agreement ceiling and a 3.8B student lands on statistically identical numbers offline and closed-loop (§6.3). On the calibrated grid the two scales sit within ±0.38% of each other on delay at 95% confidence, a bounded difference rather than a bare non-rejection, and across the wider sweep reproduce each other's control trajectory exactly on a tenth of all cells (§6.7). The bottleneck is not capability but the distillation ceiling.

3. **A large, seed-unanimous fine-tuning effect and the measurement condition needed to see it**: on a Bloomsbury grid at its default demand, where under 15% of vehicles clear, distillation is worth under two points. Calibrated so that 87% clear, the distilled controller beats the fixed-time floor where the stock controller loses to it, an 18.3% head-to-head improvement holding on 30 of 30 seeds. Oversaturation compressed the effect on this network. The same calibration on the corridor reproduces neither the size nor the significance of that effect, so I report a single-network result with its failed replication (§6.5).

4. **The authorship finding**: on saturated maps and behind the shield's envelope, where delay effects are compressed, offline accuracy still decouples from closed-loop delay, but fine-tuning raises the SLM's share of logged decision ticks, from 0.39–0.62 to 0.56–0.77, and cuts its divergence from MaxPressure between two and fourteen-fold. The complement is the anti-starvation floor, which fires on 24 to 43% of ticks with no SLM at all, so what fine-tuning removes is the stock proposer's excess above that rate rather than the floor itself. Authorship moves even where delay does not. I report this with the rival imitation reading stated (§6.6).

5. **A replication failure reported against my own claim**: a three-seed sweep showed MaxPressure worse than naive fixed-time on every real London topology. At thirty paired seeds it is significantly *better* on Old Street, tied on Bloomsbury, and significantly worse on Euston, on relative delay though not on absolute. I retract the claim in favour of a narrower one: MaxPressure's advantage over a naive plan is unreliable on irregular real geometry, though uniform on synthetic grids (§6.1).

6. **A proven audit-to-law pipeline**: seven collision scenarios reconstructed from signed, hash-chained records, with tampering caught and the governing UK rule inferred deterministically from verified facts into cited evidence packs rather than verdicts (§6.8).

7. **A scoped trust mechanism**: asymmetric, discount-only trust whose ledger, replayed against each run's recorded claims, catches most blatant misreporting and is evaded by a calibrated stealth attacker. Gating was off in these runs, so the ledger scores what it would have caught, not what it prevented. Run against fine-tuned as well as stock controllers, it also suggests that authorship may cut both ways, a blatant attack tending to cost the fine-tuned arms more than stock, in four of six controller-map cells at three seeds (§6.9).

8. **Two negative results under fair protocols**: first, in an authorisation re-test, explicit in-prompt policy rescues a stock model's recall but not its safety, with the fine-tuned students format-captured on the adjacent task (§6.11); second, serving-artifact provenance alone shifts a headline delay result by nine points, which is why every comparison uses an identical-pipeline control (§6.2).

1.4 Scope and thesis statement

The locked thesis question is: *is an on-device Foundry-Local SLM traffic controller that matches or beats classical control on real London topology, while producing a provable accident audit, possible at all? And at what model scale and configuration?* This is a feasibility and scale-threshold study: everything runs in SUMO on three real London topologies plus synthetic grids, only Euston demand magnitude is measured, realisations are synthetic throughout, and the comparators are classical, not trained deep-RL, since the target is feasibility rather than a new state of the art in delay.

In answering it I argue something narrower than feasibility. Inside a deterministic shield, what fine-tuning reliably moves is *authorship*, not delay. Stock students already author 39 to 62 per cent of logged decision ticks, a majority on four of the five cells; the fine-tuned students author more, on every cell I tested, though the size of that shift varies widely and I did not isolate its mechanism. Delay behaved differently. It moved materially only on the one network I calibrated to clear, and not on every network that clears: the corridor I calibrated did not follow. That gap is the thesis. An accident audit concerns the certified model only so far as the model authored the decisions, so authorship is the property that has to hold. Chapter 6 tests both halves separately, against both classical comparators. Chapter 7 reports where each half holds, and Chapter 8 says where it fails.

1.5 Dissertation structure

Chapter 2 positions the work, and Chapters 3–5 give the architecture, the engineering and the evaluation protocol. Chapter 6 reports the evidence in the order it was built, and Chapters 7–8 answer the research questions.

Chapter 2

Background and related work

2.1 Classical signal control and its guarantees

Deployed urban signal control rests on two families. Fixed-time control serves a cycle engineered offline from historical counts, optimising nothing online but deterministic, starvation-free, and indifferent to sensing noise. Its adaptive rival MaxPressure [27], the canonical strong baseline, serves at each intersection the phase of greatest "pressure" (upstream minus downstream queues, weighted by turn ratios), a decentralised rule Varaiya proved throughput-optimal, stabilising every stabilisable demand. That guarantee is bought with assumptions which, read against the backpressure literature, fail structurally rather than marginally in central London.

First, the optimality proof assumes unbounded queue storage. Gregoire et al. [7] prove that linear-pressure control of Varaiya's form is not work-conserving in the general case once queue capacities are finite, and construct explicit deadlocks; their repair, capacity-normalised pressure, is *not* part of standard MaxPressure. Central London's short links imply small capacities, so that failure condition is plausibly standing rather than exceptional here, though I do not measure it. Second, the only stability result for a deployable (non-oracle) backpressure controller holds under heavy load alone: Gregoire et al. [6] require every queue to exceed saturation flow, concede that sub-saturation queues "can unstabilize the queuing network", and call stability outside it "still a challenging problem". Measured diurnal demand sits below that threshold most of the day; one candidate mechanism there is the pressure signal chasing arrival noise where a fixed cycle holds steady, which I do not test. Third, every favourable empirical figure here is a grid figure: roughly 90% of oracle backpressure on a uniform 21×21 grid, about 80% under heterogeneous parameters, and the authors caution that grid results cannot extend to arbitrary networks. Fourth, junctions with give-way movements, gap acceptance, or gyratory circulation fall outside the model domain entirely: service in these theorems is binary and phase-determined, so no guarantee applies.

The field's own tables corroborate this: AttendLight [20] reports MaxPressure *losing* to naive fixed-time by 4.6–18.6% on some T-junctions and two to four times on its two four-way layouts with unequal lane counts (INT10, INT11): the collapse is geometry-specific. Against this stands a scope boundary: at least six independent studies [8, 14, 29, 32, 33, 36] show MaxPressure beating fixed-time by 7–58% on grid-like CityFlow and SUMO networks, including nominally "real" Manhattan, Jinan, and Hangzhou topologies, all on synthetic or replayed-uniform demand. My measurements (§6.1) fall between those two bodies of evidence: the sign changes with the geometry, while my synthetic grids behave conventionally (n=3). That pre-empts the objection that my baseline is broken, and stops short of the stronger reversal an earlier three-seed sweep had suggested. From a targeted rather than systematic review, I find no prior work empirically testing MaxPressure against fixed-time on measured UK geometry under measured demand, still less connecting the outcome to the theorems above.

2.2 Learning-based signal control

Deep reinforcement learning dominates the literature classical control left behind: phase-competition invariances in FRAP [35], graph-attention coordination in CoLight [29], cross-layout transfer in AttendLight [20], all surveyed by Wei et al. [28]. The results are strong on the benchmarks the community chose, almost uniformly CityFlow grids with synthetic or replayed demand, typically single-run, so headline numbers carry no variance estimates. Two further limits motivated the turn to language models: trained policies are opaque, giving no account of *why* a phase was served, and the sim-to-real gap goes unexamined in an evaluation monoculture of shared simulators and demand models. This dissertation does not compete as an RL method: it inherits the framing (phase selection on a fixed interval) but rejects the evaluation habits.

2.3 LLMs and SLMs for traffic signal control

LLMLight/LightGPT [14], the closest prior work, distils GPT-4 rationales into LoRA-tuned students down to 0.5B. Yet it serves them on four A800-80GB GPUs and evaluates single-run on CityFlow, with no seeds or confidence intervals. Traffic-R1 [36] trains a 3B model with GRPO, earning credit for a ten-intersection production deployment and a comparison of fine-tuning paradigms showing SFT degrading the base model's general benchmarks. Its own future work concedes edge deployment remains unrealised, the artefact this dissertation builds. DGLight [32] (a dissertation preprint, not peer-reviewed) tunes an 8B model on dual H100s and calibrates the field: the LLM-versus-MaxPressure travel-time gap it and its peers report spans 1.1–4.2%, a margin narrower than the seed variance I measure on the corridor topologies and that no competitor tests for. CoLLMLight [33] scales to 196 intersections on server GPUs and finds its fine-tuned 8B model beating a stock 671B model, a single data point for distillation over scale that motivated my central bet. CuraLight [8] is the one fine-tuning work on SUMO, but its 12B model needs roughly 44GB of VRAM against my 6GB budget. iLLM-TSC [21] inverts my safety topology on a single synthetic four-way intersection: an RL agent proposes and a cloud GPT-4 vetoes, with a retry loop falling back *silently* to the unvetted RL action. Its degraded-communications threat model is a scenario axis I do not test. LATS [34] distils an LLM teacher into MLP/GRU students, so nothing generative is deployed, and its pure-LLM baselines perform worst of its methods, corroborating my finding that stock small models are weak deciders.

The pattern across all seven is consistent: every system serves from cloud or server GPUs, every headline table is single-run or at best $n \leq 10$ without statistical tests, none pairs control with any audit mechanism, and none evaluates on measured UK topologies.

2.4 Small language models, distillation, and quantisation

I adopt the SLM survey [19] and on-device review [30] framing: this system is *edge-only* (no cloud fallback), reported against the review's indicator template (time-to-first-token, VRAM, latency percentiles). Neither survey contains a traffic or control-loop case study, so the application gap exists by omission. The nearest on-device comparator, Octopus [2], is a 2B function-calling model at 1.1–1.7s on Android, bracketing the 0.94–1.64s median decision latency of my deployed students. The surveys corroborate a reliability floor: Qwen2.5's own report [23] shows instruction-following (IFEval) collapsing from 84.1 at 72B to 27.9 at 0.5B, consistent with the 45–69% decision accuracy of my stock 0.6B model. The Qwen3 report [24] complicates this. Its 0.6B model posts respectable IFEval scores yet was the weakest decider of the Qwen3 family I served, parsing 93–100% strict JSON while choosing wrong phases, which sharpens the point: generic instruction-following benchmarks do not measure numeric decision quality, and the technical reports never test it.

On distillation there is a tension: Orca [18] argues small models need rich explanation traces, whereas my distillation is label-only. Three defences. First, the task is a bounded, closed decision schema closer to classification than open generation, for which the knowledge-distillation survey [31] endorses labelling-plus-SFT. Second, Orca itself concedes small models excel in constrained settings. Third, my student nears the teacher ceiling (97.7% against 97.3–99.3% inter-model label agreement on 600 cross-checked items). I concede what Orca implies: out-of-distribution generalisation and natively generated audit rationales are untested risks. The jump from 55% to 97.7% has precedent in distilled students beating data-scale expectations [10, 26]. LoRA [11] and QLoRA [4] justify rank 16, though their evidence stops above 125M parameters, and I flag the extrapolation.

Quantisation is the second and larger tension. QLoRA trains adapters on a 4-bit base and serves unmerged, whereas I train QLoRA-style, merge to bf16, then apply post-hoc int4 round-to-nearest. That is two quantisation steps, and QLoRA's authority does not transfer to the second. The literature predicts trouble there: GPTQ [5] shows small models are the hard case for round-to-nearest, AWQ [17] finds the gap narrowing at 4-bit with grouping, and post-hoc quantisation after fine-tuning has been measured losing 9.4% relative even with GPTQ [25]. My contrary finding is therefore scoped narrowly: equal aggregate held-out decision accuracy, not output identity, under weight-only int4, on a near-converged narrow-schema checkpoint, measured with a discrete accuracy metric coarser than perplexity. Finally, forgetting-scaling results [13] predict general-capability drift at my very low training loss, so I carry this as a limitation (§7.3), which Traffic-R1's fine-tuning-paradigm result independently motivates. What no published work supplies is per-decision reliability data for a generative model at 0.6B, int4-quantised, inside a closed control loop.

2.5 Safe-by-construction shields

The canonical formalism is shielding [1]: a reactive system synthesised from an LTL safety specification and a finite MDP abstraction by solving a two-player safety game, placed either preemptively or post-posed. It carries proofs my system does not: correctness for all abstraction-consistent environments, minimal interference, and preserved learner convergence. The authors also warn the shield must remain active after learning, which mine does. Two weaker rungs sit below it. Invalid-action masking [12] proves the masked update is a valid policy gradient, shows masking scales where penalties fail, and finds agents trained masked remain largely valid unmasked. Neurosymbolic masking [9] learns the symbolic grounding when constraints are unknown, and its own future work points back towards temporal-logic specifications.

My shield is *not* a shield in the Alshiekh sense: it composes three weaker but sound pieces, enforced by code rather than proved from a specification (§3.3). There is accordingly no minimal-interference proof and no lookahead: my floor reacts *at* the starvation threshold where a synthesised shield acts before violation becomes unavoidable. The upgrade path is cheap at single-junction scale, and I name it as future work. My topology inverts iLLM-TSC's (§2.3, §3.1), which a 6GB edge budget makes viable; I did not implement the inverse to measure the difference. No prior LLM-based signal-control work states the formal status of its safety layer at all.

2.6 Audit, transparency, and legal grounding

My headline thread, that an edge controller should produce a provable account of its decisions, sits in a three-way gap in the trust literature. On the attack side, Qu et al. [22] show roughly 6% of vehicles feeding falsified data to an undefended learning controller: the colluders cut their own waiting time by 92.5% and raise everyone else's by 62.7%. Their most dangerous result is that a *calibrated* lie beats a maximal one, which motivates my stealth-liar scenario; their paper names real-time anomaly detection as unbuilt future work. On the integrity side, the blockchain architecture of [16] rejects 100% of spoofed inputs at ~39ms but never names a signature scheme, and its own future work concedes failure when validators collude. Third and most fundamentally, input integrity is not decision accountability: it certifies what the controller *heard*, not what it *decided* or why.

My audit design therefore borrows from outside traffic: Certificate Transparency [15] shows that an append-only Merkle log with inclusion and consistency proofs makes tampering detectable without trusting the log operator, the property a post-accident investigation needs. Its per-decision cost, a signature and a hash, fits the edge budget where a validator network does not; my chain inherits the property only once its head is externally witnessed (§3.4). The legal layer is modest: Dahl et al. [3] profile pervasive hallucination when language models answer legal questions, which rules out letting any model assert fault, least of all a 0.6B one. My system instead emits a *cited evidence pack* (§3.4) against a UK-law knowledge base in which authority-side liability is legally weak and arises essentially only from conflicting-green misfeasance, driver-fault doctrines are strong, and emergency-vehicle exemptions are conditional. I am aware of no prior signal-control system that signs and chains individual control decisions, or that mounts a defence-side trust mechanism, though I claim this from a targeted rather than systematic review; building both yields the crash-to-UK-law evidence pack of §6.8.

Chapter 3

System design

A learned controller at a junction must be *optional for safety and mandatory for accountability*: the deterministic control path must be complete without the model, while every actuation must be provable afterwards, whoever authored it. This chapter presents the architecture realising that premise and defines the three structural quantities (authored share, proposal acceptance, divergence rate) whose measurement becomes the dissertation's central finding.

3.1 Architecture overview

Each junction runs an identical control loop on a fixed decision interval (≈10 s simulated). Per tick, local state (per-phase queue and waiting measurements from the junction's own sensors) is rendered into a prompt, and the on-device SLM served by Foundry Local proposes a phase that a deterministic parser extracts and the safety shield (§3.3) masks, validates, and possibly substitutes. The model is not consulted on every tick: a congestion gate serves MaxPressure whenever total halting demand falls below a threshold (two vehicles by default), intended to keep the learned component for the states where it can matter; I ran no gate-disabled arm at evaluation power. The served phase actuates the lights, and a signed audit record joins the run's hash chain under the junction's key (§3.4). MaxPressure is computed every tick regardless of the SLM and is at once the shield's reference, its fallback, and the strong baseline, so the learned component costs one model call and its marginal *safety* risk is bounded by construction. Its delay risk is not: a stock proposer behind the same shield loses 22.7% to MaxPressure on the calibrated grid.

Three consequences follow. First, graceful degradation is the default: a model timeout, garbage output, or a dead runtime (all observed, §6.12) leaves the junction running MaxPressure with the failure recorded. Second, the architecture inverts iLLM-TSC [21] (§2.5): the LLM proposes and a local $O(\text{phases})$ verifier vetoes, since that path must be cheap, local and always available; I did not implement the inverse. Third, with the deterministic path complete, the SLM's value is measurable as a *difference* against the same loop with the proposer disabled.

3.2 The SLM decision agent

The agent contract is minimal: given a system instruction and a rendered state, the model must return exactly `{"phase": N}`. Temperature is zero, the serving endpoint OpenAI-compatible, and parsing falls through three tiers in the control loop (braced JSON, keyed, bare integer). The offline evaluator adds a strict-JSON tier and records the tier per response, so format discipline is measurable, not assumed.

Four prompt formats act as graded capability probes. **Myopic** exposes only per-phase halting counts, a least privileged view answerable by a counting heuristic. **Sota** adds delay-aware context (accumulated waiting), the state a delay-optimal policy needs. **Prediction** adds short-horizon arrival forecasts, probing anticipatory information. **Coordination** adds neighbour notes, probing multi-junction awareness. The same rendering code serves both offline evaluation and closed-loop control, so formats are identical and both measure the same model on the same inputs. Whether offline accuracy predicts loop outcome is itself a result (§6.4).

3.3 The safety shield

The shield composes three mechanisms, named with the formal literature's vocabulary (§2.5):

1. **Preemptive action masking** [12]: the model chooses from a fixed menu of conflict-free phases, so no proposal can create a conflicting green.

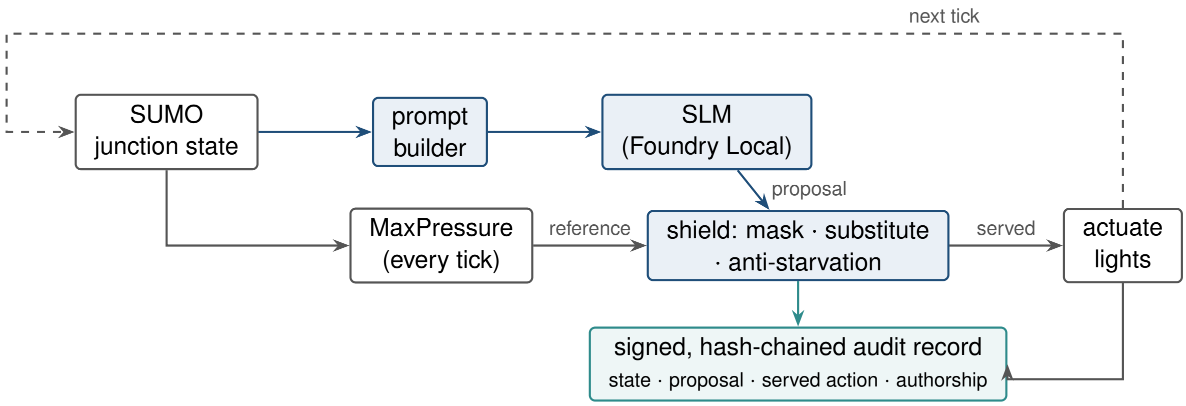

Figure 3.1: The junction control loop. The deterministic path (grey: state → MaxPressure → shield → actuation) is complete without the SLM; the learned proposer (blue) attaches to it, and every served decision lands in the signed audit chain (teal), whoever authored it.

2. **Post-posed default substitution**: an invalid, unparseable, or absent proposal is replaced by the MaxPressure phase and recorded as a served decision the SLM did not author.

3. **Anti-starvation runtime monitor**: a minimum-service floor force-serves any green phase passed over for more than a bounded number of consecutive intervals, overriding both the SLM and MaxPressure, except while an admissible emergency preemption holds the signal. It is best-effort rather than proved: only one phase can be served per interval, so when two reach the threshold together the code records the residue as a violation.

This is a sound heuristic composite, not a correct-by-construction shield in the Alshiekh sense [1]: no synthesis from a temporal-logic specification, no minimal-interference proof, a limitation with an upgrade path (§7.4). It still enforces, for every arm including the baselines, no conflicting greens and no approach starved beyond the floor except where the code records a violation.

Three measured quantities are defined on this structure, and their denominators matter. Every logged decision tick is labelled by *who authored the phase actually served*: SLM, MaxPressure substitution, or anti-starvation floor.

Authored share is slm/decisions, the quantity the audit argument needs: the fraction of *logged decision ticks* whose phase originated with the SLM. The denominator is the ticks the controller logged, those where the congestion gate admitted the SLM plus gate-skipped ticks on which the anti-starvation floor changed the served phase; intervals the gate skipped without a phase change are served by MaxPressure and counted on neither side. The ratio therefore understates neither arm, but is a share of *decisions taken*, not of wall-clock signal time.

Proposal acceptance is slm_served/slm_valid_proposals: of the ticks where the SLM returned a well-formed proposal, the fraction where that proposal was served rather than displaced. In practice the anti-starvation floor displaces it.

Divergence rate is diverged_and_served/slm_served: of the decisions the SLM authored, the fraction whose phase differed from what MaxPressure would have chosen. It measures how far the served policy departs from the classical reference, *not* how often the shield vetoed the model. Masking makes such a veto structurally impossible for unsafe actions, and in the fine-tuned and stock 0.6B arms substitution fires on a negligible fraction of ticks (§6.6).

None of the three metrics counts unsafe actions blocked: the substitution path catches only malformed or absent output. Reporting all three separately prevents the interpretive error of reading a high acceptance rate as a high share of authorship (§6.6).

3.4 Audit trail and crash reconstruction

The audit layer is designed backwards from what an investigator asks after a collision: *what did the controller know, what did it decide, who authored it, and can the record be trusted?*

Every logged decision tick appends a record containing the halting-queue state vector, the model's parsed proposal, the shield's proposal, the served action, authorship (SLM / shield-substitution / anti-starvation) and the override flag; model identity and per-call latency are recorded once at run level, not per tick. Records are Ed25519-signed per junction and hash-chained into a single append-only chain per run. Chain and signature verification run end-to-end after every experiment, and the threat model includes an imposter re-signing attack (caught in all evaluated scenarios, §6.8). The chain head can be anchored to external witnesses in a Certificate-Transparency-style quorum [15], with a permissioned-ledger anchor implemented as one optional witness rather than as a contribution.

On top of verified records sits the reconstruction layer: a UK-traffic-law knowledge base (statutes, case law, the conditional emergency-vehicle exemption) against which a collision scenario's verified facts are matched deterministically. The output is a *cited evidence pack*, never a verdict: the governing rules, the verified facts triggering them, and the fault anchor each rule implies. That answers the legal-hallucination risk [3] of §2.6: the legal knowledge is a curated, citable database and the adjudicative act is left to humans. My scenarios use synthetic actors on real topology, so no real data subject exists, and the residual legal exposure of a real deployment is analysed in §7.3.

The audit's value depends in turn on authorship (§3.3). An evidence pack over a deployment whose decisions are mostly floor overrides attributes the behaviour to a fairness timer, not to the learned component. The audit is about the model in proportion to what the model authored, and fine-tuning raises that share (§6.6).

3.5 Trust-coefficient mechanism

Coordination requires acting on neighbour reports, which a compromised neighbour can abuse. The trust mechanism prices that risk with three commitments. First, **local sensing is the ultimate truth**: a junction's own sensors are never doubted, and a neighbour claim is scored against what local sensing later verifies, within a claim lifetime after which it is provisionally discounted. Second, **asymmetric update**: verified truths earn trust slowly (prior 0.5, ten consecutive truths to reach 0.9) while a detected lie from a high-trust source collapses it in one step (lie factor 0.25), because in infrastructure trusting a liar costs more than doubting an honest peer. Third, **discount-only wiring**: trust can strip a caught liar's power to corroborate a preemption, but can never *add* an action on zero local evidence. Discounting alone cannot make safety worse, and a real ambulance within sensing range is preempted on the junction's own detection, never on a neighbour's trust. A truthful report verified late is vindicated retroactively, so honesty is never permanently penalised for latency. The adversarial evaluation and its partial results are reported in §6.9.

3.6 The fine-tuning axis

The final axis asks what changes when the proposer is fine-tuned rather than stock, and its pipeline is label distillation: a frontier teacher labels junction states sampled from training-reserved seeds, a second frontier model re-labels four audit samples of the teacher's output (600 states, 98.5% agreement, §4.3), and a deterministic sanity verifier screens the teacher's own answer rather than the state, rejecting a label that serves an empty phase while other approaches are waiting, or that names a phase outside the junction's range. The teacher is therefore second-guessed on exactly two degenerate cases, never by a weaker rule competing on the decision itself. The four prompt formats are rendered from three separately sampled and labelled state pools into one mixed training set. Training is QLoRA [4], and the merged model is compiled to weight-only int4 ONNX and served through the same runtime as every stock model (§4.3, §4.4).

Three rules protect the comparisons, enforced where specified (§5.3, §5.4): a holdout hash-frozen before any training, closed-loop evaluation seeds disjoint from training seeds, and a control arm of the same base model compiled through the identical int4 pipeline, so the only manipulated variable is the weights on the 0.6B axis. The 3.8B student has no identically-compiled stock control. Ablation students test memorisation and specialisation hypotheses: leave-one-topology-out and four format specialists, all trained under the same recipe.

Chapter 4

Implementation

This chapter records the engineering that made the design measurable: the simulated London networks, the on-device serving stack and its documented failure modes, the distillation and compile chain, and the experiment harness. I keep it tight and evidence-focused, with Appendix A tabulating the reproducibility pins (versions, seeds, templates).

4.1 Simulation environment and topologies

All control runs in SUMO with TraCI. Six topologies span a regular-synthetic to irregular-real axis along which Chapter 6 finds the largest differences, without isolating which property carries them: two synthetic grids (3×3, 4×4), a Bloomsbury grid from OpenStreetMap, the Euston A501 corridor and its peak-hour variant, and the Old Street complex junction. Euston's AADF-derived demand carries the A501's measured DfT vehicle mix but an assumed hourly profile, with the peak-hour variant rescaled to the measured DfT peak-hour magnitude. Old Street's nine signals include a single five-arm, seventeen-movement junction, demand-calibrated after default random demand gridlocked it. The Bloomsbury grid and Euston peak-hour corridor got the same treatment (§6.5). Only Euston's demand magnitude is measured, the other two real maps being randomTrips draws (§7.3).

4.2 On-device serving with Foundry Local

Serving is Microsoft Foundry Local (pinned 0.8.119) on an RTX 2060 with 6 GB VRAM, consumer hardware of roughly signal-cabinet class and a hard project constraint: it caps the ladder at ~4B (7B-class models OOM the runtime, recorded), which is what makes the scale-saturation result (§6.3) deployment-relevant rather than academic.

Custom models serve through two execution providers. The CPU path works out of the box, but WebGPU requires editing the compiled model's `genai_config.json` provider options and delivers about 1.6× lower latency at *equal aggregate holdout accuracy* on the one artifact and format where both providers were scored (§6.3), which licensed running every sweep GPU-served. The comparison is within one artifact, not a claim of absolute speed: my pipeline-compiled 0.6B artifacts serve at roughly 2.6 times the median latency of the vendor catalogue build of the same weights (§6.12), a real cost of controlling the compile chain. Windows ML CUDA autoregistration requires a newer Windows build than this machine's, making WebGPU the effective GPU path.

Three reliability findings from sustained unattended serving are mitigated in harness (§4.5) and reported in §6.12.

4.3 Distillation dataset and training

From eight training-reserved seeds ($\{3, 7, 11, 29, 42, 57, 91, 123\}$) I sampled 48,137 unique junction states across three separately sampled pools: 16,988 delay-aware states that also render the privileged myopic rows, 18,596 prediction and 12,553 coordination. A frontier teacher labelled them, and a second frontier model re-labelled 600 of them in four audit samples of 150, agreeing on 98.5% (97.3–99.3% per sample). Rendering the four prompt formats over those pools yields 59,533 training examples and a hash-frozen holdout of 5,580 (`HOLDOUT_MOD`=12). Rendering is family-aware: ChatML with an empty think block and no-think marker for Qwen3, Qwen2.5's ChatML without the marker, and Phi's `<|system|>...<|end|>` convention. Loss applies to the completion only, targeting exactly `{"phase": N}`.

Training is QLoRA r=16, NF4 double quantisation, fp16 compute. The 0.6B student trained on the serving machine itself (13.5 h on the RTX 2060), so the full loop from teacher to served controller fits on one consumer device. The 3.8B student and its five ablation variants trained on a single UCL RTX 4090 (2.3 h each, 8.9M trainable parameters, 0.23%).

4.4 Compile-and-serve chain

Six steps take merged bf16 weights to a served endpoint: QLoRA merge → int4 RTN compile → config patch → manifest synthesis → cache registration → WebGPU provider swap. That chain is, to my knowledge, the first documented end-to-end recipe for serving a custom fine-tuned model through Foundry Local; Appendix A records each step with the failure that established it.

4.5 Experiment harness

Two harnesses produced the closed-loop sweep and offline-evaluation numbers. The **factorial runner** executes model × format × topology × seed cells with per-cell raw JSON artifacts, a pre-flight serving probe per cell, and multi-pass retry for deferred cells; the **offline evaluator** scores holdout rows against the served endpoint. Standalone probes supplied the rest: the fixed-time floor, emergency preemption, the attack battery, the crash-law audit, the authorisation re-test and the retention probe. Both harnesses are fail-loud, resumable and recoverable in the senses Appendix A defines. They survived a six-week unattended campaign including two service restarts, one machine reboot, and one mid-write power loss (one corrupted cell, detected by JSON validation and re-run).

Chapter 5

Experimental methodology

This chapter defines what I measured, how each comparison was controlled, and the statistical decision rules I committed to before running the experiments. The stance throughout is fail-loud: harness errors abort or are counted, never silently scored, and null results are reported as findings.

5.1 Evaluation axes and metrics

I evaluate along four axes, each with a primary metric.

Closed-loop control quality. The primary metric is mean network delay: the mean, over all vehicles that *departed* into the network, of total time in network (SUMO's tripinfo `duration`), over a fixed 1,200-step horizon. A completed trip contributes its full travel time and a vehicle still running at the horizon the time it has accrued. Every controller cell is paired with a fixed-time baseline on the *same topology and seed*, so the reported quantity is the paired relative delay change. The secondary metrics are completed-vehicle throughput, teleport count and undeparted backlog, which guard against delay "improved" by starving demand.

Offline decision quality. Accuracy against frozen frontier-teacher labels on a held-out set of junction states, at int4 precision exactly as served. I also report strict-JSON rate (the fraction of responses matching `{"phase": N}` exactly) and parse rate (any recoverable phase), because failing the output contract costs differently from choosing a wrong phase.

Structural authorship. Authored share, proposal acceptance and divergence rate, defined in §3.3, reported alongside the decision composition itself (SLM / substitution / anti-starvation).

Serving viability. Decision latency (p50/p99) against the 10 s decision interval, and a log of serving-reliability events with their mitigations.

5.2 Statistical protocol

Every powered closed-loop cell is replicated over n=30 seeds per topology; the exploratory sweep that preceded them ran three seeds per cell and is labelled as such throughout. The test for a delay effect is a one-sample t-test on the per-seed relative change against each baseline, computed on the same quantity as the reported CI, and I pool across topologies (n=90) where the hypothesis is topology-independent. The explanatory study's ANOVA uses the project's pure-mathematics regularised incomplete beta, so no library version can silently change it. The closed-loop t and Wilcoxon p-values are SciPy's, recomputable from the committed per-seed JSON. For the factorial explanatory analysis I report effect sizes (partial η^2) alongside F statistics, and for headline comparisons I state direction, magnitude and p together rather than significance alone.

Two commitments were made in advance. First, the honest-null policy: a non-significant result at n=30 is reported with its magnitude and uncertainty, not suppressed or re-run until significant. Second, no mid-experiment stopping: cell counts were fixed by the factorial design before launch, and the resumable runner (§4.5) re-runs only cells that produced no measurement (crash, probe failure), never cells whose answer was unwelcome.

5.3 Seed-disjointness and holdout freezing

Two leakage paths exist between training and evaluation. The first is closed by construction; the second is narrowed, not closed.

State-level leakage. The distillation dataset is built from junction states sampled on eight reserved seeds (Appendix A). The offline holdout is split from those pools by a deterministic hash of the state content, frozen at build time, before any training example was rendered. No training example, under any format rendering, shares a state with a holdout example.

Trajectory-level leakage. Closed-loop evaluation uses thirty seeds fully disjoint from the training seeds: $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$. This matters because a controller fine-tuned on states from seed s has seen a compressed replay of that seed's trajectory, so evaluating on it would overstate generalisation. The route files are static, with fixed vehicle identities and departure times, and the SUMO seed varies microscopic car-following and lane-change randomness rather than the demand itself. Disjointness therefore narrows this path without closing it: a held-out seed is a different realisation of the same demand, not a different demand. Head-to-head arm comparisons are unaffected, since both arms run the identical route file and seeds, but generalisation beyond the trained demand is not established here. Three families do not use the disjoint set. The first two, the powered base-model factorial of §6.2, reported over seeds $\{1..30\}$, and the exploratory sweep that motivated the powered design, at three seeds per cell, involve no model fine-tuned in this project, so seed disjointness carries no leakage meaning for them. The third, the trust battery of §6.9, does run fine-tuned controllers on training-state seeds, and is flagged there. Every other closed-loop *controller* number in Chapter 6 is a disjoint-seed number.

5.4 Baselines and controls

Fixed-time is the floor every deployed junction achieves: each network's own default cyclic plan, the same phase menu as every other arm. **MaxPressure** is the theoretically-grounded adaptive baseline [27], and doubles as the shield's fallback policy.

The control for every fine-tuning comparison is the *same base model compiled through the identical int4 pipeline* (denoted stock, or ft0), not the vendor catalogue artifact. The choice has two motivations. First, I observed the catalogue Qwen3-0.6B GPU artifact emitting corrupted tokens after long service uptime, an unlogged operational observation rather than a measured finding (§6.12). Second, and independently, that artifact differs from my compiled one in quantisation recipe even when healthy. The second reason is sufficient on its own: holding the compilation pipeline fixed leaves the fine-tuned weights as the single manipulated variable.

One model class is excluded by measurement rather than swept: phi-4-mini-reasoning's p99 decision latency (11.1 s) exceeds the 10 s decision interval on this hardware, so no deployment could serve it in time. That is a deployability finding, not a control one: the harness blocks on the model call with simulated time stopped, so a closed-loop cell would flatter a slow model rather than penalise it. The p99 is also one format of four on one seed, the other three at 8.69, 9.61 and 9.74 s. It is recorded as latency-excluded, its single corridor probe retained descriptively.

Two further controls target specific hypotheses. A **leave-one-topology-out** (LOTO) student, trained with all Old Street states removed, tests whether distillation gains are topology memorisation, and four **format-specialist** students, each trained on one prompt format alone, test whether a deployed junction would prefer a per-format model over a generalist. The answer (§6.3) was no, at the single scale (3.8B) that sweep ran at.

5.5 Experiment inventory

Table 5.1 maps every experiment family in Chapter 6 to its design and artifact trail. All raw cells, per-seed records and aggregation scripts are retained in the repository, so every number is recomputable.

The full campaign spanned three machines and roughly six weeks of wall-clock compute.

Table 5.1: Experiment inventory.

Family	Design	Primary output
Base-model factorial	8 catalogue SLMs × 4 formats × 6 topologies × up to 32 seeds	closed-loop delay vs baselines
Explanatory (why) study	two-way ANOVA, 2 models × 4 topologies × 3 seeds (config fixed)	F, η^2, interaction
Teacher cross-audit	second frontier model re-labels 4 batches of 150 teacher-labelled states	label agreement
Offline distillation	7 phi rows + 4 qwen rows × 4 formats, frozen holdout	accuracy, JSON rate
Fine-tune closed-loop	ft vs stock at 0.6B, plus a 3.8B ft confirmation arm, × 3 topologies × 30 disjoint seeds	paired delay, authorship
Decision battery	cloud frontier vs local SLM on identical states	agreement, latency
Accident / crash-law suite	scripted collision scenarios → audit reconstruction	evidence packs, chain verification
Trust battery	honest / blatant / stealth misreporting	detection, delay impact
Emergency / EV suite	scripted EV arrival and spoofing attack on the arterial corridor, 3 modes × 30 paired seeds	clearance metrics

Chapter 6

Results

Results follow the order the evidence was built: the baseline landscape first, then the demand calibration the throughput guards forced, which yields the headline result, then authorship, forensics and the audit, trust and reliability batteries.

6.1 Baseline landscape: what the fixed-time floor revealed

In the exploratory sweep (three seeds per cell, daily-resolution demand) the theoretically throughput-optimal MaxPressure [27] appeared *worse than naive fixed-time on every real London topology* (Euston +15%, Old Street +5%, Bloomsbury +3%) yet dominated the synthetic grids (−27% on grid4×4, −19% on grid3×3). Replication kills it: Table 6.1 recomputes it paired per seed over the thirty training-disjoint evaluation seeds.

Table 6.1: MaxPressure vs fixed-time, paired per seed (n=30). Positive = MaxPressure worse. CI is 95% on the mean relative change and p is the one-sample t on that same quantity, the convention used throughout this chapter.

Topology	MaxPressure (s)	fixed-time (s)	rel. (%)	95% CI	p
Euston peak-hour	263.8	233.0	22.3	[+5.6, +39.1]	**.011**
Bloomsbury grid	529.0	528.6	0.2	[−1.6, +2.0]	.82
Old Street	146.6	161.0	−8.2	[−12.3, −4.2]	$\mathbf{2.6{\times}10^{-4}}$
Bloomsbury calibrated	158.4	173.7	−8.8	[−9.2, −8.3]	$\mathbf{8{\times}10^{-27}}$
Euston calibrated	151.7	155.4	14.2	[−2.6, +31.0]	.095

At n=30 the picture is not a reversal but a **sign flip that tracks geometry**: on Old Street, the study's most complex geometry at five arms and seventeen movements, MaxPressure is *significantly better* than fixed-time, contradicting the exploratory result outright, while on the Bloomsbury grid at its default oversaturated demand the two are indistinguishable. But on the same network calibrated to clear, MaxPressure is 8.8% better, so every "tied on Bloomsbury" statement here is scoped to the oversaturated regime. Calibration does not rescue it everywhere, but neither does it settle the corridor: there the same rule leaves the comparison inconclusive (p=.095, a CI straddling zero, and absolute means that favour MaxPressure by 2.4%).[1] The same treatment therefore produces different outcomes on the two networks, which leaves saturation unconfirmed as a *sufficient* account of MaxPressure's weakness without identifying what replaces it on the corridor: the two networks differ in several respects at once and nothing here varies them independently. On the saturated Euston corridor before calibration MaxPressure is *significantly worse* than fixed-time too, with a caveat: its per-seed spread is heavily skewed, and on absolute rather than relative delays the paired test would not reach significance (p=.083), making Euston the least robust of the three real networks to how the question is posed.

A third classical baseline tests whether any of this is specific to *pressure*. Longest-queue-first serves the largest upstream queue with no downstream term, so unlike MaxPressure it will discharge into a blocked exit, though the two are routinely conflated. I ran it over the same thirty seeds with shield, minimum green, clearance and anti-starvation floor identical between the arms. **The downstream term buys nothing measurable on these networks**: longest-queue-first is statistically indistinguishable from MaxPressure on all five topologies (−0.9% to +13.4%, every $p \geq .30$), and it reproduces MaxPressure's pattern against the naive floor, worse on the corridor (+13.6%, p=.027) and better on Old Street and the calibrated

[1] The +14.2% is the mean of per-seed ratios, the convention used throughout this chapter, and it is inflated by a single +140.3% seed. On absolute paired delays the same comparison gives p=.82. The row is reported because the convention is applied uniformly, not because it supports a direction.

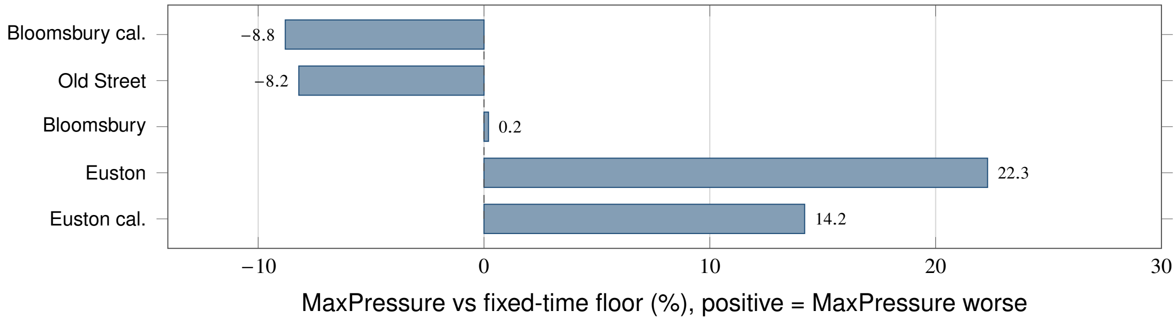

Figure 6.1: The classical baseline is not uniformly strong. MaxPressure is significantly worse than a naive fixed-time plan on the Euston corridor, significantly better on Old Street, and tied on the Bloomsbury grid. Its advantage tracks both geometry and saturation, which is why every controller claim in this dissertation is stated against the fixed-time floor with MaxPressure reported alongside.

grid (−5.9%, −8.3%). Whatever makes the classical policy weak on a saturated arterial, the pressure formulation is not detectably part of it, though the intervals are wide (Euston calibrated [−12.7, +39.5]), making this a non-rejection rather than a demonstrated equivalence. It weakens one candidate without removing it; saturation, link storage and the assumed turning split all remain.

MaxPressure's advantage is uniform and substantial only on the regular synthetic grids the literature favours, which are the cells replication did not reach: those figures remain three-seed. Backpressure theory suggests why the arterial is its weakest case, as the unbounded-storage and heavy-load assumptions of §2.1 [6, 7] are both violated by a saturated short-link corridor: a hypothesis consistent with the Euston direction, not a demonstrated explanation, since nothing here manipulates storage capacity to test it. Either way, SLM-vs-MaxPressure comparisons on real topologies are confounded by MaxPressure's own variable strength, so every controller claim is stated against the fixed-time floor, with MaxPressure alongside.

6.2 Base-model factorial

The base-model sweep (eight servable catalogue models × four prompt formats × six topologies) established three facts before any fine-tuning. It is a sweep, not a full factorial: three models were never swept on the later measured-demand corridor, phi-4-mini-reasoning is latency-excluded (§5.4) after a single-seed corridor probe, and the ANOVA is fitted on the balanced subset. The 6 GB card caps the ladder at ~4B (§4.2) and a runtime schema change blocked a newer generation.

First, in that exploratory sweep the best shielded stock arm beat the fixed-time floor on every topology, from −4.3% on Bloomsbury to −29% on grid4×4 (Old Street −14%, seed-noisy). Those are per-map bests, not the class: averaged over all arms the sweep is worse than the floor on Bloomsbury. The per-map win does *not* survive replication on the measured-demand corridor, so it is motivation for the powered design below, not evidence. Second, results must be conditioned on topology: pooling across maps is a Simpson's-paradox trap. Third, across base models per-decision reasoning quality does not predict closed-loop value: on 80 identical junction states cloud frontier models agree with the delay-optimal heuristic at 92–96% against 51–72% for local SLMs, yet a near-chance local model still won closed-loop cells. Within one base it can (§6.5). The controller's value is plausibly systemic: SLM, shield and gating together, though no arm here removes the shield or the gate to test that, and this battery is open-loop by construction.

An explanatory two-way ANOVA (Table 5.1) confirms the structure: topology dominates (partial η^2=0.97), the model matters (partial η^2=0.83), and the two interact strongly (F=32.2, p<0.001, partial η^2=0.86), though at three seeds per cell this is exploratory replication, not the powered design.

The powered stock-model replication (n=30 per cell, seeds 1–30, four catalogue models × two formats, both baselines) sharpens this per topology. Stock models have no training set, so the training-disjoint

Table 6.2: Holdout accuracy (%) against frozen frontier-teacher labels, int4-as-served. All fine-tuned rows and the stock Phi row emit 100% strict JSON; stock Qwen3-0.6B reaches 93.4% on sota.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1	63.3	69.3
Qwen3-0.6B generalist ft	**87.9**	**97.7**	**97.6**	**99.0**
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini generalist ft	**88.7**	**97.1**	**97.4**	**99.3**
Phi-4-mini LOTO	88.3	96.8	97.4	99.1
Phi-4-mini sota-specialist	87.9	97.8	89.5	93.3
Phi-4-mini myopic-specialist	87.0	87.9	88.8	91.8
Phi-4-mini prediction-specialist	87.9	91.3	97.3	98.4
Phi-4-mini coordination-specialist	87.4	89.1	94.7	99.3

seeds used elsewhere are not required here. On **Old Street** stock SLM control achieves the campaign's strongest powered wins against the fixed-time floor: Qwen2.5-0.5B sota −11.8% (p<0.001), Qwen3-0.6B sota −9.6% (p<0.001), Phi-4-mini myopic −8.6% (p<0.001), with one arm also significantly beating MaxPressure (Qwen2.5-0.5B sota, −4.9%, p=.028).

On **Euston peak-hour**, with demand set to the DfT-measured peak-hour magnitude, directional split and vehicle mix for the A501 count points (turning proportions assumed, single weekday survey), the result reverses: every stock arm is **worse than the fixed-time floor by** 7.4% **to** 19.7%, significantly on three of eight arms (two-sided p .017–.026, a fourth at p=.053), yet statistically indistinguishable from MaxPressure. Not a null but a measured harm. The powered run's measured peak-hour magnitude loads the corridor *less* heavily than the daily average it replaces: 780 vehicles against 1,071, 44.3% completing against 33.1%. Whether that lighter load is what makes the harm visible cannot be said here, because §6.5 finds that reducing Euston's saturation further does not behave as a saturation-compression account predicts. **Distillation does not repair this.** Against the same comparator the fine-tuned Qwen and Phi arms are +15.5% and +16.2% here (Table 6.3), inside rather than below the stock band. An earlier draft claimed the opposite, an artefact of scoring fine-tuned arms against MaxPressure and stock arms against fixed-time. Whether the deficit is topological or a residue of congestion that calibration could not remove is left open in §6.5.

The replication also yielded an unplanned methodological finding. On the 26 seeds shared between runs, the *catalogue* Qwen3-0.6B artifact and my identically-prompted, identically-shielded compile of the same weights differ by nine points of mean relative delay (+4.1% vs −5.2% against MaxPressure) on this high-variance topology: quantisation and packaging provenance alone shift the headline sign. The gap is comparator-robust though the sign flip is not, because against the fixed-time floor the same two artifacts are +17.4% and +4.8%, both worse than the floor but still 12.6 points apart. So §5.4 runs an identical-pipeline control, and so should the field, because "the same model" is not the same controller across serving artifacts, on two of the three topologies where I could compare them (Old Street −0.2%, p=.93).

6.3 Offline distillation: saturation at 0.6B

Table 6.2 gives the complete offline campaign: both student scales, stock versus fine-tuned through the identical int4 pipeline, plus LOTO and format-specialist controls, on the frozen 5,580-state holdout as served (WebGPU, int4).

One naming point comes first, because two Qwen checkpoints appear in this dissertation and they are not the same model. The row above is the *generalist* student, trained and evaluated on all four formats, while every closed-loop result in this chapter uses the *sota-only* student, trained on the delay-aware format alone and served in it, which scores 97.4% against the generalist's 97.7% on the only format either is asked to act

Figure 6.2: Offline decision accuracy against frozen teacher labels, at int4 precision as served. Distillation closes almost the whole gap, and the two fine-tuned students land on each other at every format: six times the parameters buy nothing per-decision. The myopic format is the exception at 88%, which is the privileged-teacher ceiling described in §6.3.

in. The offline table therefore characterises the distillation rather than the provenance of the closed-loop arms.

Five findings follow.

1. **Scale saturation.** The 0.6B and 3.8B generalists are statistically indistinguishable on every format: the largest gap, 0.8 points on myopic, sits against a standard error of about 1.2 points on n=1,510, so no two-proportion test approaches significance. That test is conservative here: both students are scored on the same frozen items, so the comparison is paired, but I retained only aggregate accuracies per format, making the exact paired test (McNemar) uncomputable after the fact. Both students sit at the label-noise ceiling on three of four formats (97 to 99% against 97.3 to 99.3% agreement between the two frontier models that produced the labels, on 600 cross-checked items), myopic excepted at 88% and explained below. Six times the parameters buy no measurable per-decision gain.

2. **Where the gain lands depends on the base.** The 0.6B is lifted by +29.7 to +42.6 points on every format, while Phi-4-mini gains most on the delay-aware sota format (+38.1) and negligibly on myopic (+1.2): for a capable base, fine-tuning raises decision quality from a floor already high on three of four formats (87.5, 59.0, 86.0, 87.0) rather than instilling format discipline, since stock Phi is already 100% format-compliant.

3. **Generalists only.** No format specialist beats the generalist on its own format by more than 0.7 points while off-format costs run from 0.8 to 9.2 points, and the myopic specialist loses *even on-format* (87.0 vs 88.7), consistent with mixed-format training regularising against the noisy privileged-teacher labels that pure-myopic training overfits. The specialist sweep ran at 3.8B only, a single-scale result, and the sota-only Qwen student used in the closed-loop runs is not part of it.

4. **No detectable topology memorisation.** The LOTO student, trained with all Old Street states removed, matches the generalist on every format. Old Street is 4.2% of the sota rows and 0.4% of coordination, so this bounds a held-out-topology deficit near 7 points rather than measuring one; per-topology accuracy was not retained.

5. **Quantised serving costs no measurable accuracy here.** The CPU and WebGPU int4 artifacts score identically in aggregate (97.42% both on the sota holdout, n=1,510). Per-item outputs were not retained, so this is equality of accuracy, not demonstrated output identity, and the "lossless" claim is scoped narrowly in §7.3.

The myopic ceiling (~88%) is most likely intrinsic to the format rather than a training failure: the teacher was privileged with waiting-time state the myopic prompt hides, an Orca-style label-only distillation cost

Table 6.3: Closed-loop delay against both classical comparators, and decision composition (n=30 disjoint seeds per cell). Negative Δ favours the controller; significance is in Table 6.4. *auth* = share of all actuations authored by the SLM; *a-s* = share forced by the anti-starvation floor; *acc* = proposal acceptance; *div* = divergence from MaxPressure among SLM-authored decisions. The substitution path is in Table 6.7.

Topology	Arm	vs fixed-time	vs MaxPr.	auth	a-s	acc	div
Euston peak-hour	ft	+15.5%	−3.9%	0.64	0.36	0.70	0.12
	stock	+7.6%	−6.2%	0.62	0.38	0.68	0.46
Bloomsbury grid	ft	+0.8%	+0.7%	0.70	0.29	0.73	0.17
	stock	+2.8%	+2.6%	0.62	0.38	0.65	0.46
Old Street	ft	**−11.6**%	−3.1%	0.56	0.44	0.90	0.11
	stock	**−9.7**%	−1.1%	0.39	0.61	0.54	0.60
Pooled (n=90)	ft	+1.6%	−2.1%				
	stock	+0.2%	−1.6%				
Euston	phi ft	+16.2%	−3.8%	0.64	0.36	0.70	0.13
Bloomsbury	phi ft	+0.8%	+0.6%	0.70	0.30	0.73	0.17
Old Street	phi ft	**−9.7**%	−1.0%	0.57	0.43	0.88	0.12
Pooled (n=90)	phi ft	+2.4%	−1.4%				

[18]. Every myopic arm bar the stock 0.6B, fine-tuned or not, lands within 1.7 points of that ceiling, consistent with a label-noise floor. The non-privileged-label control that would establish it was not run.

6.4 Closed-loop outcome: delay decouples from accuracy

Table 6.3 reports the Qwen3-0.6B closed-loop sweep: fine-tuned (ft) versus identically-compiled stock, three topologies × 30 training-disjoint seeds, paired per-seed against *both* classical comparators (§6.1).

The comparator decides the outcome. Against the fixed-time floor every arm significantly beats it on Old Street and none does on the other two maps. Against MaxPressure the only arm in this table to reach significance is stock on the Bloomsbury grid (+2.6%, p=1.3×10^{-3}), a harm and not a gain. But one catalogue stock arm does beat MaxPressure on Old Street (§6.2). The calibrated regime changes both pictures and is reported in §6.5. Comparator-invariant is the negative that matters: a +42-point offline accuracy gain produces no consistent delay change in these three regimes, and the pooled effect is indistinguishable from zero under either comparator. The decision battery anticipated this **delay-decoupling paradox**. Fine-tuning's one significant effect is protective: stock control significantly *harms* the congested Bloomsbury grid where the fine-tuned controller is neutral, the head-to-head favouring fine-tuning by −1.8% (p=.022). The mechanism is taken up in §7.2: masking confines both arms to one conflict-free menu and the anti-starvation floor serves 23–61% of ticks, though the floor's own share varies with the arm (§6.6), so much served behaviour is identical by construction whatever the proposal quality. The substitution path is not part of this. I ran no unmasked arm, so this is a design-level reading, not a measurement. The headroom may also be small to begin with: external calibration puts the whole LLM-vs-MaxPressure gap at 1–2% [32]. Because paired t-tests on means hide the skew the catastrophic tail of §6.7 puts into the per-seed distributions, Table 6.4 adds 95% CIs and a Wilcoxon signed-rank test on the *typical* seed.

The Old Street win holds on the typical seed as well as the mean ($p \leq .002$ both tests). The Euston intervals are wide and zero-straddling on the topology with the heaviest failure tail, so the characterisation is *typical-case behaviour with a catastrophic tail*, pointing at tail control (a per-decision timeout, or an outlier detector reverting to the classical policy) rather than at better average decisions.

These rank-test results are not confirmatory: of roughly forty paired comparisons across the dissertation, the two reaching $p < .05$ only under the rank test would not survive a Holm correction across that family,

Table 6.4: Closed-loop effects against the fixed-time floor, conventions as in Table 6.1, plus a Wilcoxon signed-rank test on the paired per-seed delays (n=30). Bold marks $p < .05$. The MaxPressure counterpart of every row is in Table 6.3.

Arm	Topology	rel. (95% CI)	p_t	p_{Wilcoxon}
Qwen ft	Euston	+15.5 [−1.9, +32.9]	.079	.309
	Bloomsbury	+0.8 [−1.0, +2.6]	.354	.516
	Old Street	**−11.6** [−17.2, −6.0]	**.0002**	**3.2e-6**
Qwen stock	Euston	+7.6 [−4.8, +20.0]	.222	.349
	Bloomsbury	**+2.8** [+0.9, +4.6]	**.005**	**.008**
	Old Street	**−9.7** [−13.3, −6.1]	**5.1e-6**	**1.2e-6**
Phi ft	Euston	+16.2 [−1.4, +33.9]	.070	.245
	Bloomsbury	+0.8 [−0.9, +2.5]	.358	.543
	Old Street	**−9.7** [−15.4, −3.9]	**.002**	**7.1e-5**

and I pre-registered no confirmatory comparison. The fine-tuned-versus-stock head-to-head is nonetheless the study's primary contrast by design, the only cell in which a single variable is manipulated, and at p=2.0×10^{-21} it survives Holm where they do not.

The Phi-4-mini confirmation arm (final rows of Table 6.3) extends saturation end-to-end. The 3.8B fine-tuned student's closed-loop profile is not distinguishable from the 0.6B's on either comparator, with the same divergence band (0.11 to 0.17) and authorship profile; on the calibrated grid the intervals bound any difference within ±0.4% of delay. It serves at the lowest latency of my compiled arms (Table 6.9). On the scale-threshold question that is a bounded non-difference, not a demonstrated equivalence (§6.5).

6.5 Throughput guards, and the result the gridlock was hiding

Mean delay can be improved by stranding vehicles, so §5.1 commits to guard metrics alongside it. Table 6.5 changes how two of the three topologies read.

Table 6.5: Guard metrics, means over the same n=30 training-disjoint seeds as every delay comparison. "Compl." is completed vehicles as a share of those loaded within the 1,200-step horizon; teleports are SUMO's gridlock-escape removals.

Topology	Arm	Loaded	Compl.	Teleports	Undep.
Euston peak-hour	fixed-time	780	51.1%	22.2	184
	MaxPressure	780	45.3%	31.1	218
	qwen ft	780	48.7%	28.1	200
	qwen stock	780	50.3%	23.3	199
Bloomsbury grid	fixed-time	1715	15.1%	115.0	328
	MaxPressure	1715	14.6%	119.3	336
	qwen ft	1715	14.2%	120.0	341
	qwen stock	1715	13.5%	135.6	348
Old Street	fixed-time	480	83.3%	0.7	1
	MaxPressure	480	85.8%	1.3	0
	qwen ft	480	86.7%	0.5	0
	qwen stock	480	84.4%	1.0	1

Of the three original cells, only Old Street is measured in a free-flowing regime: it clears, teleports are effectively absent, mean delay is well-posed, and every arm significantly beats the fixed-time floor there (Table 6.4). Euston sits between at 45–51% completion, where the SLM arms *complete more vehicles than MaxPressure* (380 and 392 against 353) without a significant delay advantage, though not more than the fixed-time floor's 399; no significance test was run on completion. **Uncalibrated Bloomsbury is a gridlock-regime measurement**: under 15% complete, over a hundred teleported out of deadlock, so its mean delay compares two congested-failure modes. The stock arm completes fewer vehicles and teleports more than MaxPressure, so the stock harm survives the guard test in direction, but is regime-scoped.

Completion also reproduces §6.1's ordering (MaxPressure better on Old Street, tied on Bloomsbury, worse on Euston) without using delay at all, so without the skew I caveated there, the tie appearing here as MaxPressure marginally worse (14.6% against 15.1%). **The sign flip is therefore not an artefact of a fragile mean**, though it does not identify the network property responsible: completion is as consistent with saturation or link storage as with geometry.

Bloomsbury carried the study's only significant closed-loop result against MaxPressure, so I re-examined it. Two diagnostics identified the regime. Tripling the horizon to 3,600 steps raised completion only to 35% while multiplying teleports roughly sevenfold (121 to 830), so the network was over capacity, not under-simulated. Subsampled demand located the boundary sharply, clearing at 40% (87.6% completion, zero teleports) and collapsing again at 55% (57.5%). I therefore added a demand-calibrated variant at 40% (686 vehicles) and re-ran all three arms over the identical thirty seeds, retaining the oversaturated cells.

Is this a principled correction or a search for a more favourable test bed? In favour: the guard metrics were recorded in every cell from the campaign's first run, so their values were not produced by looking; the diagnosis is outcome-independent; and the 40% target matched Old Street's completion rate, not any effect size. Against: the decision to look followed a weak result, a data-dependent choice that retaining the superseded cells does not by itself redeem. Uniform application is the check I can offer: Euston peak-hour also fails the guard and is where SLM control looks *worst* (§6.2), so calibrating the corridor risks the conclusion rather than protecting it. I calibrated it, though the probed fractions bracketed failure only on the grid.

Calibrating Euston does not rescue the corridor, and the distillation effect does not survive there at full size. The treatment worked partly: completion rises from 45–51% to 68–74%, but in the MaxPressure arm 7.7 vehicles per run still teleport and 56.9 never depart, against 0.0 and 0.0 for both classical comparators on the calibrated grid, where only the stock arm departs from that (1.70 teleports, 5.7 undeparted). On the partly cleared corridor the fine-tuned arm is still +8.5% against the fixed-time floor ($[-2.4, +19.5]$, p=.12) and the stock arm +34.8% ($[+5.3, +64.4]$, p=.023). The mean spares the fine-tuned arm only because the tail is enormous: it beats the floor on six of thirty seeds, its median seed is +11.7%, and the rank test rejects (p=.016). Neither arm differs significantly from MaxPressure (−0.3% at p=.93, +21.9% at p=.062). The head-to-head favours fine-tuning in direction and on the typical seed: median −7.4%, 23 of 30 seeds, Wilcoxon p=.045. Its mean, though, is −3.7% and not significant ($[-19.1, +11.7]$, p=.63), across per-seed outcomes from −71.6% to +104.5%.

Two conclusions follow. First, and against me: **Saturation explains the Bloomsbury result; on Euston it is reduced, not tested**. The same correction that transformed the grid's marginal result yields, on the corridor, a fifth of the magnitude it produced there and no significant mean, though in absolute delay (−27.4 s against −35.9 s) the two are comparable. I cannot attribute the surviving deficit to topology: nothing here varies geometry while holding demand and storage fixed. Topology and residual saturation both stay live, and the grid effect did not replicate on the second network I calibrated. Second, the threshold was applied where it could damage the conclusion, and did: the rule, take the largest demand fraction that still clears, produced an unfavourable result, reported as such. The rule is stated here, not encoded in the calibration script, which prints completion at each probed fraction and leaves the choice to the analyst.

The gridlock was masking the effect, not creating it. On the calibrated network the stock controller degrades mean delay by **11.9% against the fixed-time floor** and **22.7% against MaxPressure**, an order of magnitude beyond the +2.8% it costs oversaturated, while both fine-tuned students **beat the fixed-time floor by 8.8%**, within 0.03% of MaxPressure. The head-to-head is the largest, most certain effect in this dissertation: 18.3% less delay than the identically-compiled stock model ($[-19.8, -16.8]$, $p=2.0\times10^{-21}$), **winning 30 of 30 seeds**, the Phi student reproducing it (−18.3%, 30/30, $p=8.2\times10^{-22}$). The thirty seeds

Table 6.6: The same controllers under two demand regimes on the same Bloomsbury network (n=30 paired seeds each). Positive Δ is worse than the fixed-time floor.

Regime	Arm	Δ vs fixed-time	Δ vs MaxPressure	Compl.	Auth.
Oversaturated	qwen ft	+0.8 [−1.0, +2.6]	+0.7 [−0.7, +2.1]	14.2%	0.70
(14% clears)	qwen stock	**+2.8** [+0.9, +4.6]	**+2.6** [+1.1, +4.2]	13.5%	0.62
Calibrated	qwen ft	**−8.8** [−9.4, −8.2]	+0.00 [−0.56, +0.56]	87.4%	0.76
(87% clears)	phi ft	**−8.8** [−9.3, −8.3]	+0.03 [−0.43, +0.48]	87.5%	0.77
	qwen stock	**+11.9** [+10.0, +13.8]	**+22.7** [+20.6, +24.7]	76.4%	0.59

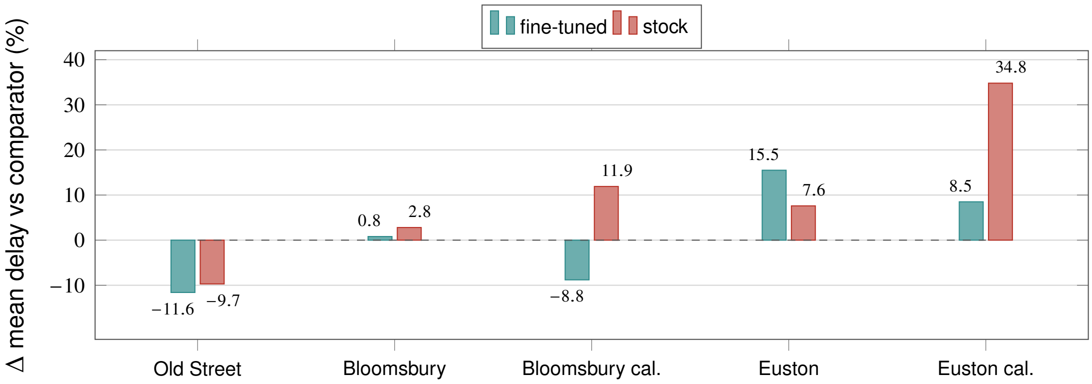

Figure 6.3: Closed-loop delay against the fixed-time floor, n=30 training-disjoint seeds per cell. Negative is better than the floor. The distillation effect is the gap between the two bars: it is largest on the calibrated Bloomsbury grid, where the fine-tuned student beats the floor by 8.8% while the stock model loses to it by 11.9%, and it does not reproduce at that size on the calibrated corridor.

vary SUMO's microscopic noise over one fixed 40% demand draw, so they replicate within a scenario rather than across demand. The mechanism is the one §7.2 develops. Throughput corroborates it (stock 76.4% against the fine-tuned 87.4%), and the stock arm shows no serving pathology (484 valid proposals of 546 decisions, no fallbacks, normal latency), so the degradation is in the decisions, not the endpoint.

This also bounds the scale-saturation result instead of leaving it a non-rejection: the 3.8B and 0.6B students differ by +0.03% with a 95% CI of [−0.32, +0.38], so any difference between the scales is within ±0.38% of delay. That stops short of demonstrated equivalence, which would need a margin pre-registered against an operational threshold (a TOST procedure) this study did not define.

An oversaturated test network can hide a large effect: the original Bloomsbury cells alone would have concluded a marginal 1.8% at p=.022. On Euston the same correction did not reveal a comparable effect, so this is a demonstrated failure mode on one network of two, not a systematic bias whose size I can quantify. The literature evaluates predominantly on synthetic grids under demand rarely reported in clearance terms (§2.2), so saturation-driven compression of published effect sizes deserves attention. Reporting completion and teleport counts alongside delay costs nothing and would expose the regime, though not the compression: grid3×3 is oversaturated at 37.8% completion and still shows −19.1%.

6.6 Authorship: what fine-tuning actually moves

The structural columns of Table 6.3 carry the dissertation's central finding, which emerges only if the three denominators of §3.3 are kept apart.

Any claim that an SLM arm "matches" MaxPressure would be circular were the shield quietly substituting the classical phase. **It is not.** Across fourteen arm-by-topology cells and 420 seed-runs the substitution path fired in exactly one cell, the fine-tuned arm on the Bloomsbury grid, at 0.54% of decisions on average

Table 6.7: Decision composition for every closed-loop experiment (n=30 training-disjoint seeds per cell, sota format). Columns are shares of all logged decision ticks except *diverg.*, the share of *SLM-authored* decisions whose phase differed from MaxPressure's. The *shield* column is the substitution path, the shield replacing an invalid or absent proposal with the MaxPressure phase; in the MaxPressure arm there is no SLM to substitute for, so the column does not apply and the classical policy authors every non-floor tick.

Topology	Arm	SLM	shield	floor	diverg.
Euston peak-hour	MaxPressure	0.00	n/a	34.1	n/a
	qwen ft	0.64	0.00	35.7	0.12
	qwen stock	0.62	0.00	38.2	0.46
	phi ft	0.64	0.00	36.0	0.13
Bloomsbury grid	MaxPressure	0.00	n/a	28.4	n/a
	qwen ft	0.70	**0.54**	29.5	0.17
	qwen stock	0.62	0.00	37.6	0.46
	phi ft	0.70	0.00	29.6	0.17
Old Street	MaxPressure	0.00	n/a	43.1	n/a
	qwen ft	0.56	0.00	43.6	0.11
	qwen stock	0.39	0.00	61.2	0.60
	phi ft	0.57	0.00	43.2	0.12
Bloomsbury calib.	MaxPressure	0.00	n/a	24.3	n/a
	qwen ft	0.76	0.00	23.5	0.04
	qwen stock	0.59	0.00	41.4	0.56
	phi ft	0.77	0.00	23.3	0.04
Euston calib.	MaxPressure	0.00	n/a	36.3	n/a
	qwen ft	0.62	0.00	37.7	0.07
	qwen stock	0.59	0.00	40.6	0.37

(110 in total, 8.99% on its worst single seed). Every other cell is 0.00% on every individual seed: wherever the gate admitted them, stock and fine-tuned models alike returned a well-formed, in-menu proposal (`shield_fallbacks_none` is zero in every SLM cell of the campaign).

The anti-starvation floor is the large non-SLM component, at 23 to 61% of ticks, and it authors 24 to 43% of the *MaxPressure* arm on the same maps. The rule is identical in every arm but its share is not: paired per seed, the stock arm carries 4 to 18 points more floor than MaxPressure on every map ($p \leq 1.2{\times}10^{-9}$), while the fine-tuned arm sits at the MaxPressure rate. The floor therefore intervenes most in the arm that performs worst, and what fine-tuning removes is the stock proposer's excess over the no-SLM rate.

Fine-tuning raises the SLM's true authorship of actuations, most where stock was weakest: Old Street 0.39 to 0.56, Bloomsbury 0.62 to 0.70 (0.77 once calibrated), Euston 0.62 to 0.64. The floor column is the arithmetic complement (Old Street 0.61 → 0.44), so it adds no independent evidence; proposal acceptance, which does, rises with it (0.54 → 0.90). The natural reading is that the distilled model serves neglected approaches before the fairness timer must, but the logs record only who served each tick, not which approach, so that cannot be separated from the arm simply tripping the timer less often. The honest headline is the authored share, not the acceptance rate: even the best fine-tuned arm leaves *23 to 44% of actuations authored by a hard-coded timer*.

Cutting against the simple reading, **the fine-tuned arms land level with MaxPressure while they choose what MaxPressure would have chosen**: divergence among SLM-authored decisions falls from 0.46–0.60 to 0.11–0.17, and on the calibrated networks that carry the headline it falls to 0.04 and 0.07, the two lowest in the study, with authorship at 0.59 against 0.76–0.77. That is genuine authorship of a near-classical policy, not a scaffold masquerading as one, but the deflationary reading is strongest exactly where the delay result

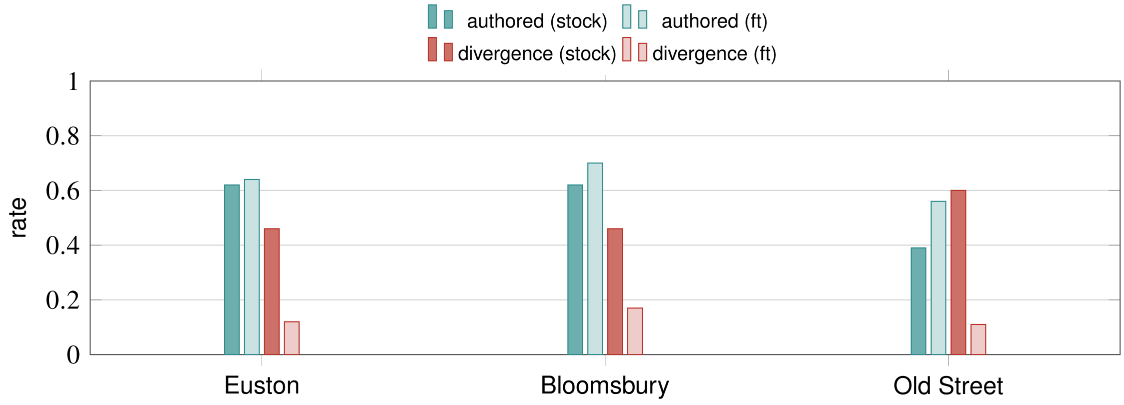

Figure 6.4: What fine-tuning moves (n=30 seeds per bar). Teal: the SLM's share of *logged decision ticks*. Red: divergence from MaxPressure among SLM-authored decisions.

is largest. Distillation may have taught the student to imitate the classical reference, an explanation neither established nor excluded, but two observations bound it. First, the teacher was a frontier model reasoning over accumulated waiting, not MaxPressure, and the students agree with its held-out labels at 97–99% (§6.3), so any convergence is at most indirect. Second, closed-loop delay cannot discriminate imitation from rediscovery: against the fixed-time comparator MaxPressure is +22.3% on Euston peak-hour against the fine-tuned arm's +15.5%, similarity rather than separation, and on the calibrated grid the two coincide within 0.03%. Only the remaining one-in-ten divergences, where the arms differ from both baselines, could discriminate them, and I have not characterised those, so the most substantive open question here needs a decision-level counterfactual study I did not run.

What survives is the audit-relevant claim at its true magnitude: over the stock Old Street deployment an accident audit could attribute 39% of actuations to the learned component, over the fine-tuned one 56%, the balance falling to a deterministic timer, a smaller claim than *the model is the controller*.

6.7 Failure forensics: the audit layer explains the tail

The audit architecture's value shows most clearly when control fails. Of the 270 uncalibrated-topology closed-loop cells, 14 (5.2%) are catastrophic (delay $\geq$ +20% vs the seed-matched MaxPressure baseline), and all 14 signed chains verify, so each failure has a provable record.

First, **failures are a controller property, not seed hardness**: the catastrophic seeds are baseline-easy or median demand realisations (baseline delay ranks 1–20 of ~35, never the hard tail), and no seed breaks all three arms.

Second, the two failure regimes have different, attributable mechanisms. Four of the stock arm's five failures are its own served decisions: in the worst cell (Euston s13, +110.3%) the chain shows 214 of 335 decisions SLM-authored, 119 divergent proposals served (override 0.556), zero shield fallbacks and normal anti-starvation, so the collapse was the model's policy. Same-seed fine-tuned arms sit at +1.1% and +5.5%. The fifth (Euston s19) sits at arm norms and the chain clears it.

Third, **the two fine-tuned students converge at the trajectory level on a substantial minority of cells**. Across the 90 shared cells, nine produce *bit-identical* mean delay, 25 more agree within 0.5 points, and 56 differ. The identical cells are separate runs serving the same phase sequence, not duplicated records: wall-clock differs by up to 2.15× and median latency by 2.23×, and in one both arms took 307 decisions with SLM/floor attribution differing by exactly one tick, where floor and model chose the same phase. Four of the five catastrophic fine-tuned cells fall inside this converged subset, which restates the convergence by construction rather than explaining it; the independent evidence is the decision-level agreement above. I read it as the behavioural counterpart of offline saturation, in which the distillation target constrains not only what the students score but what they *do*. The limit stands: partial convergence is equally consistent with both students defaulting alike on easier cells, and I have not separated those explanations.

Counted across the calibrated regimes, the same catastrophic cells show the clearest effect fine-tuning has anywhere in this study, and it is not an effect on mean delay. Paired over the same thirty seeds against the identically-compiled stock control, catastrophic cells fall from **20 to 0** on the calibrated Bloomsbury grid and from **8 to 1** on the calibrated Euston corridor, with the 3.8B student reproducing the grid result at 0. Where the network cannot clear the tail does not move: 3 to 3 on Euston peak-hour, 2 to 1 on Old Street, 0 to 0 on the gridlocked grid. **Distillation removes the harm tail on both networks that clear**, including the corridor where it moved no mean delay, so the tail and the mean answer to the same clearance condition but not to the same degree. I did not isolate the mechanism.

One honest gap: this campaign persisted aggregate decision statistics per cell, not per-step audit records, so *when* within a run the divergence concentrated is unrecoverable here, though the per-step chain is retained in the dedicated accident-suite runs (§6.8). Full tables in `results/failure_forensics.md`.

6.8 Audit and crash reconstruction

The audit layer was evaluated on seven pre-defined collision scenarios spanning the legally distinct cases (red-light running, EV exemption, maintenance vehicle, amber gambling, conflicting green, dark signal, spoofed EV). All seven prove end-to-end: hash chain verified, signatures verified, and an imposter re-signing attack caught in every scenario. The governing UK rule is *inferred deterministically from the verified facts*: spoofed-EV correctly denies the exemption and anchors fault on the driver, and conflicting green alone attributes (weak, misfeasance-only) authority fault. The output is a cited evidence pack for a human adjudicator, never a verdict [3].

Above the per-run chain sits the Certificate-Transparency-style completeness layer of §2.6. Over a twelve-entry batch, a salted Merkle root hides the entry contents while remaining provable, the salt is erasable off-chain, and an inclusion proof verifies while the same proof under a *forged* salt is rejected. Under a two-witness cross-audit the witnesses agree and the batch is consistent. When one witness is made to serve a different root, **the equivocation is detected and consistency fails**, and a witness refuses to overwrite its own history. The cross-auditor's own honesty is an **unverified assumption**, on a par with the identity root. One further limit: the mechanism passes but the batch is **not externally anchored**. I withheld the public-log write rather than pollute a production transparency log.

6.8.1 A worked evidence pack

Figure 6.5 traces the hardest of the seven end to end: a collision at a junction that emitted a conflicting green.

Each cited rule carries the fact that triggered it, so an investigator can audit the inference rather than read a conclusion, and the doctrine cited runs against the deployer.

6.9 Trust-coefficient battery

The trust mechanism passes its full property battery (9/9): a persistently honest junction climbs to 0.929 trust, a persistent liar collapses to 0.011 and loses the power to corroborate, the designed asymmetry holds in both directions, and a truthful-but-late report is vindicated retroactively by local sensing (§3.5).

In closed-loop traffic (honest, blatant and stealth misreporting injected into SUMO) I ran the battery against three controller arms: fine-tuned 3.8B, stock 0.6B and fine-tuned 0.6B. One scoping point governs every number below: trust gating is off in these runs, so the ledger did not alter any served phase. Detection is scored by replaying each run's recorded claims and local-sensing confirmations through the ledger, which measures what the mechanism would have caught, not what it prevented. The delay costs reported here are therefore the attack's effect on an undefended controller. No stock-3.8B trust run exists, so the stock/fine-tuned contrast is available at 0.6B only. It is the fine-tuned deployment that matters (§6.6), since

```
VERIFIED RECORD (seq 0, junction J, hash 93dc0a0b..., t=100.0)
  chain verified: True   signatures verified: True
  imposter re-sign attempt: CAUGHT
VERIFIED FACTS
  signal_state = green      conflicting_green = True
  crossing_class = civilian second_party_on_green = False
  key_present = False       corroborated = False
GOVERNING RULES (inferred from the facts above)
  LR-authority-misfeasance [Bird v Pearce]  fault: low  (common law)
    "the audit shows the system emitted a conflicting green
     (a positively-wrong indication) -- misfeasance, not omission"
  LR-authority-nonfeasance [Gorringe v Calderdale] fault: low (common law)
    "authority fault otherwise defaults to zero; it turns non-zero only
     on this misfeasance/conflicting-green branch (low confidence;
     legal_causation NOT_ASSESSED)"
OUTPUT: cited evidence pack for a human adjudicator -- not an adjudication
```

Figure 6.5: The conflicting-green evidence pack, reproduced from `results/experiment_crash_law.json` (record, facts, rule ids, fault weights and both `why` strings verbatim; headers and closing line editorial): the only one of the seven scenarios in which authority fault is non-zero, and the case where the system's own record is used against its operator.

a controller acting on its coordination inputs is what poisoned inputs can move. Three findings, at three seeds per cell (descriptive). First, **detection is controller-independent by design and in measurement**: every controller catches and collapses a blatant liar on Bloomsbury (62–70 catches, final trust 0.21–0.27) and grid4×4 (146–151 catches, trust ≤0.03, corroboration stripped), while a calibrated stealth attacker evades essentially everywhere (≤2 catches, trust ≥0.92), an open limitation. On the corridor, Euston peak-hour and Old Street the liar junction is never consulted, so the mechanism is inert on all three (zero lies injectable). Second, **the delay cost of blatant lying survives detection**: on the synthetic grids it adds +2.7 to +11.5 points over the same controller's honest run even as the liar's trust collapses, because detection bounds future influence without undoing congestion. On Bloomsbury the damage stays within ~3 points (fine-tuned qwen −1.3% under attack against −1.1% honest). The design predicts this is the local-sensing override, but no override-ablated arm was run, so map structure is an equally live explanation and the two are not separated. Those trust figures are not comparable cell-for-cell with Table 6.3: this battery runs three seeds (42, 123, 31) in the coordination format against the same-seed MaxPressure baseline, where the main sweep uses thirty disjoint seeds in the sota format. Two of its three seeds (42 and 123) are training-state seeds and the third (31) is a disjoint evaluation seed (§5.3), so this is the one family in the study whose fine-tuned arms partly see states the students were distilled from. Third, **stealth's harm is positional, not congestive**: its intermittent inflations cost −2.3 to +2.9 points, buying retained corroboration power at high trust to abuse later. The threat model for a fine-tuned deployment is therefore asymmetric. Distributional detection is the right next defence (§7.4).

6.10 Emergency preemption: what the priority path is worth

The authorisation machinery serves a concrete function, measured over thirty seeds on the synthetic arterial corridor with a scripted ambulance and spoofing attack, in three modes: *nopreempt* (no signal priority, as at an installation without emergency-vehicle detection), *naive* (preempts on any claim), and *defended* (preempts only on a claim with a valid authorising key and corroboration, §3.5).

All four intervals exclude zero. **Signal priority is worth about half a minute to an emergency vehicle** against a no-priority controller, **security has a measurable price**, and **that price buys quantified protection**. Whether that exchange is worth making is a policy judgement. Scope: one scripted ambulance and one attack window per run, so these are per-event magnitudes, not expected values under a naturalistic call rate, and the baseline characterises no specific deployed system.

Table 6.8: Emergency-vehicle priority, n=30 seeds, 95% CIs. Positive values are seconds saved.

Quantity	Mean	95% CI
Ambulance time saved by priority (nopreempt − defended)	31.1 s	[25.4, 36.5]
Cost to the ambulance of the authorisation gate (defended − naive)	8.0 s	[3.7, 12.5]
Cross-traffic harm avoided under attack, mean	4.6 s	[3.3, 6.0]
Cross-traffic harm avoided under attack, worst case	16.1 s	[11.5, 20.4]

6.11 The fair re-test of the SLM authorisation negative

An early measured negative carried a supervisor-posed objection (Stott). The raw SLM lost to a deterministic template on its legal-note job, citation-F1 0.12 retrieval-free and 0.34 with retrieval against the template's 0.48 over 97 held-out notes, at 37.8 s per note. That gold is policy-base-derived rather than human-graded, so the negative is a pilot proxy. The objection: the model was guessing because the policy was never in front of it, so the negative cannot stand until the SLM is re-tested with explicit in-prompt rules and reason-then-classify structure. I ran that re-test on the balanced 26-claim authorisation dataset, four models × two arms (policy-scaffolded vs bare), scored against the deterministic knowledge-base gate.

The objection is partially vindicated: the scaffold works for a capable stock model. Stock Phi-4-mini rises from 42.3% to 69.2% class accuracy and stops denying genuine emergencies, its false-negative rate falling to zero. But the negative holds: even that best cell makes **four dangerous false grants** (granting preemption to claims that must be denied, four of 18 such claims) where the gate makes none in 26/26. Policy grounding rescues recall, not safety.

The re-test also produced the clearest evidence of specialisation cost: the *fine-tuned* students are far worse than their own stock bases, with ft-Phi falling to 19.2% scaffolded and 15 of 26 responses unparseable, and ft-Qwen, the generalist Qwen student (§6.3) rather than the sota-only student deployed in the closed loop, unable to emit the required `{"class", "grant"}` schema at all bare (26/26 parse failures). The harness did not retain the raw responses, so what it emitted instead cannot be shown; format capture by its training schema `{"phase": N}` is the natural reading, but an inference and not a measurement. The retention probe's mild MMLU decline (§7.3) understates the cost on near-neighbour structured tasks.

6.12 Serving reliability at the edge

Sustained on-device serving produced three documented reliability findings. One long offline evaluation died at 5,500 of 5,580 rows on a host `OSError(28)` (no space left on device) and had to be re-run; registering builder-compiled models required undocumented synthesis of the inference manifest; and I observed the catalogue Qwen3-0.6B GPU artifact producing malformed output after extended service uptime, which I record as an unlogged operational observation rather than a measured finding and which is why §5.4 takes the pipeline-compiled artifact as the control. Each was mitigated in harness (§4.5).

Decision latency is summarised in Table 6.9. The 3.8B student is the fastest of my pipeline-compiled arms and the 0.6B students the slowest, a counter-intuitive ordering that is a property of the compiled artifacts rather than of parameter count. The ordering holds only within my own compiles: four vendor catalogue arms are faster still, including the catalogue build of the student's own weights at 0.68 s, the 2.6× compile penalty of §4.2 from the other side. One call in the 0.6B arms exceeded the 10 s decision interval, so the real-time margin is comfortable at the median and not guaranteed in the tail. This is the criterion that excludes phi-4-mini-reasoning (§5.4), which differs only in frequency: a lone outlier here against a p99 above the interval there. A deployment would need a hard timeout with deterministic fallback, which the architecture already supports.

Table 6.9: Decision latency across closed-loop cells (per-cell summaries, GPU-served, sota arms). The decision interval is 10 s.

Arm	p50 (median)	p50 range	p99 (worst cell)	worst call
Qwen3-0.6B ft	1.75 s	1.39–2.57 s	7.58 s	13.79 s
Qwen3-0.6B stock	1.74 s	1.46–2.51 s	2.97 s	9.88 s
Phi-4-mini ft	0.94 s	0.92–1.20 s	1.46 s	1.49 s

Chapter 7

Discussion

7.1 Answering the research questions

RQ1 (edge competence): qualified, and topology-dependent. At powered replication (n=30 disjoint seeds) the picture splits by map. Stock SLM control significantly beats the floor on Old Street, but is significantly *worse* on three of eight arms on the measured-demand Euston corridor, overturning a win the exploratory sweep had reported. Fine-tuning does not remove that deficit, widening it against the floor from +7.6% to +15.5%. The defensible claim is narrow: competent on complex junction geometry, harmful on a saturated arterial, and superior to the naive fixed-time plan only where the network can clear.

RQ2 (what distillation buys): delay where the network can flow, authorship everywhere, and it saturates at 0.6B. Offline, distillation is dramatic (55–69% → 97–99%) and scale-flat. On a demand-calibrated grid, and on 30 of 30 seeds against the identically-compiled stock model, distillation lifts the controller above the naive fixed-time floor and is worth 18.3% delay. The two student scales sit within ±0.38% of each other, so scale saturation here is a bounded difference rather than an absence of evidence. Calibrating the Euston corridor by the same rule leaves the head-to-head not significant: the corridor deficit survives calibration, and I cannot attribute it to topology (§6.5). Independently of delay, distillation raises *ownership*: the SLM's share of logged decision ticks rises from an already substantial 0.39–0.62. For a shielded, audited controller I argue ownership is the criterion that generalises, since it holds even where delay effects are compressed (§7.2).

The saturation runs deeper than the accuracy tables show, the two students reproducing each other's control trajectory exactly on a tenth of all cells (§6.7), so a larger second student adds less diversity than its parameter count suggests, though 56 of the 90 cells still differ and no ensemble was built or costed. Against that, distillation is not free: on an adjacent structured task the fine-tuned students are worse than their own stock bases, one unable to emit the required output schema at all (§6.11). That is acceptable for a component doing one decision type behind a deterministic gate, but unacceptable if the same weights were expected to serve the authorisation or legal-note roles.

RQ3 (provable audit): yes, as designed. Seven scenario reconstructions prove end-to-end with tampering caught (§6.8). The legally weak branches are weak *in the law*, faithfully represented rather than overstated.

RQ4 (trust): partial, and the partiality is structured. The mechanism's properties hold under all three controllers tested, because the ledger scores claims against local sensing rather than model behaviour. At three seeds per cell these are descriptive magnitudes, and the three limits of §6.9 stand. The design-relevant tendency is that a blatant attack costs *more* against fine-tuned controllers than stock, which would mean authorship cuts both ways, a model that acts on its inputs being moved further by poisoned ones. I state it as a tendency because it is not uniform: of the six controller-map cells where lies are injectable, the fine-tuned cost is larger in four and smaller in two. A trust ledger is not a sufficient defence, and this study did not run a distilled deployment without one.

7.2 Why accuracy decouples from delay

Three mechanisms plausibly explain the delay-decoupling paradox, and they differ in how well this study evidences them. First, *the shield may compress the outcome distribution*: masking confines every arm to the same conflict-free menu and the anti-starvation floor serves 23–61% of ticks, intervening most in the weaker arm (§6.6), so the gap between a 59% and a 97% proposer may be far smaller served than raw. I ran no unmasked arm, and the substitution path fires on 0.54% of decisions in one cell and 0.00% elsewhere (§6.6), so it bounds nothing. This is a design-level argument, not a measurement. Second, *the headroom was small*: external calibration from a 13× larger tuned model puts the whole LLM-vs-MaxPressure delay gap at 1–2% [32], and my own decision battery showed a near-chance proposer winning closed-loop cells, because per-decision agreement with a delay-optimal heuristic is not the trajectory-level objective. Third, and the only one measured here, *the test network must be able to express the difference*: on an oversaturated map every controller is compressed toward the same terrible mean, while the same experiment on a network that clears recovers the effect on every seed (§6.5). The second and third are therefore properties of the problem and the test network; the first is a property of the controller I did not test. The practical reading: fine-tune to own the decisions, expect both the delay gain and the removal of harm tails only where the network is not already saturated (§6.7), and never evaluate a controller on one that cannot clear.

7.3 Threats to validity

Five declared threats, with defences.

Label-only distillation (construct). I distil teacher labels, not reasoning traces. Orca-style trace distillation [18] might transfer more, and the myopic format's ceiling prices the cost as a disclosed design property (§6.3).

Demand composition (external). Only the corridor's vehicle counts come from measured DfT data, but the turning proportions at each junction remain assumed. Pressure-based control acts directly on movement demand, so every controller comparison here is conditional on that assumption, and the corridor results are plausibly the most exposed to it, the A501's short links making the through-versus-turning split decisive. Nothing in this study estimates that sensitivity.

Retention (internal). Aggressive SFT can erode general capability with scale-dependent forgetting [13]. I measured it: a seeded 200-item MMLU sample through the same endpoints shows the 3.8B student at 57.5% (stock) against 53.0% (fine-tuned), a gap consistent with mild forgetting and not significant at this sample size (z=0.91, p=.37). At 0.6B the probe is uninformative by floor effect: stock already scores at chance (26.0%). No significant general-knowledge damage was detected, though 200 items bound only large effects. That is not reassurance, because the nearer-neighbour authorisation probe (§6.11) shows both students well below their stock bases. The damage reads as a narrowing onto the trained contract, which general-knowledge benchmarks are poorly placed to detect.

Quantisation scope (construct). "Lossless" int4 is claimed strictly as equal aggregate holdout accuracy under weight-only RTN. Per-item outputs were not retained, so output identity is untested, and no general equivalence is claimed, which the quantisation literature would not support [5, 25].

Simulation and demand realism (external). Everything is SUMO. Real-map demand *magnitude* is measured, but the peak-hour fraction, the temporal profile, the directional split and the vehicle-level realisations are assumed or synthetic. Comparators exclude trained deep-RL systems. Legally, the audit scenarios use synthetic actors on real topology, so no real data subject exists, but a real deployment would face two residual exposures. First, curating which rules apply is itself a judgement about relevance, so the pack cannot be treated as a neutral readout, and I cite no authority for that proposition because I know of none. Second, the evidence pack attracts only qualified privilege, which disclosure defeats.

7.4 Limitations and future work

Five directions follow from the findings. **A decision-level counterfactual**: replaying the divergent ticks and scoring what MaxPressure would have done against what the student did is what would give the imitation-versus-rediscovery question direct evidence (§6.6). **Formal shield synthesis**: replace the heuristic composite with a shield synthesised from a temporal-logic safety specification [1], since the per-junction product game may be small enough for exact synthesis and the authorship metrics defined here would then carry minimal-interference guarantees. **Closed-loop objectives**: GRPO-style tuning against trajectory reward [32] attacks the decoupling directly, because label distillation optimises agreement rather than the trajectory, and moved delay on one network of five. **Stealth-robust trust**: detection calibrated to distributional implausibility rather than single-report bounds. **Hardware-in-the-loop**: the serving-reliability findings (§6.12) are the failure class a cabinet deployment must survive, and argue for a watchdog-supervised serving architecture.

7.4.1 Deployment vectors this architecture is suited to

The evaluation above measures the hardest case for this controller, a permanent junction already running the network's own default signal plan, so matching it buys little. Two settings invert that arithmetic, and one cost question conditions both. For each I give the evidence and the remaining work.

Temporary and roadworks signals. Portable signals at a lane closure have no engineered plan, so a controller needing *no site configuration* is worth more there than one matching a tuned plan. One design property and two supporting results transfer. By design, the phase model is derived from the live network at start-up (§3.1). Measured, the leave-one-topology-out student matched the generalist across the mixed holdout (§6.3), which is evidence against topology memorisation in aggregate rather than parity on the withheld layout; and the anti-starvation floor enforces alternation deterministically except where it records a violation (§3.3), one requirement of shuttle working. Three gaps are clear. No lane-closure layout was tested. The phase model is read once, so mid-deployment reconfiguration needs re-initialisation. Shuttle working's clearance-time constraint, the safety-critical one, belongs in the shield and is not implemented.

Emergency-vehicle priority across a corridor. Installations without emergency-vehicle detection give an approaching vehicle no priority, an omission §6.10 measures at 31.1 s per event. The substrate exists: junctions exchange Ed25519-signed reports over an MQTT adapter that places a real broker between publisher and subscriber. Capability is not a deployment claim: the bus has run in simulation and over a local broker, not a field radio under loss and latency. The scenario is scripted, with the scope §6.10 declares, and corridor-scale progressive preemption is designed for but unmeasured.

Deployment economics. A market survey of published prices (September 2026)[1] places the compute node at well under 1% of site cost. A Jetson-class module lists at roughly £315–400 and runs this study's 0.6B student at a community-measured ~50 tokens/s within a 7–25 W envelope, against £80,000–130,000 for a signalised crossing. The power figure matters more than the price: it answers the objection that my 6 GB desktop GPU misrepresents a cabinet power budget, on a published figure this study did not verify. TRL's SCOOT rollout averaged a 12.8% delay reduction, which sets the bar any deployment would have to clear. Enclosure, certification, maintenance and integration dominate hardware at these ratios, so the whole-life case needs a costing study this dissertation's data cannot supply.

[1] Vendor list prices, council committee papers and published trade estimates compiled in September 2026: NVIDIA's developer price list (July 2026), a Greater London Authority estimate for signalised crossings, TRL's SCOOT rollout evaluation, and 1NCE and UK MVNO tariffs. None originates from this study's measurements.

Chapter 8

Conclusion

This dissertation asked whether an on-device small language model, served by Microsoft Foundry Local within a 6 GB envelope, can control real London junctions competently while producing an accident audit worth trusting, and at what scale the learned component earns its place.

Feasibility is a qualified yes, and the qualifications are the substance. Shielded on-device SLM control significantly beats the fixed-time floor on the study's most complex junction geometry and, once distilled, on a demand-calibrated grid. The same controller is worse on a saturated arterial whether distilled or not, and elsewhere indistinguishable from that floor except on the oversaturated grid, where the undistilled controller is significantly worse. The classical baseline changes sign with the network, and what looked like uniform weakness on real geometry was a three-seed result that replication retracted. The whole loop, from frontier-teacher distillation through QLoRA training to int4 serving, fits on one consumer device. The 0.6B student trained on the same GPU that serves it, and there distillation saturates: within this deployable envelope a student six times larger lands on statistically indistinguishable decision accuracy, which points at the distillation ceiling rather than model capacity.

The central finding reframes what fine-tuning is *for* in a shielded controller. On the maps that could not clear, the 97%-accurate student does not significantly beat the floor, and neither does anything else inside the shield's envelope, though I ran no shield ablation to isolate that compression. Where the network *can* flow, fine-tuning moves delay itself, and does so on every one of thirty seeds (§6.5). But applying the same calibration to the arterial corridor did not reproduce it, and I report that as the failed uniformity test it is. Fine-tuning moves authorship on every topology I tested (§6.6). The audit layer proves who decided what, so authorship is what makes that audit concern the certified model rather than its fallback.

The nulls bound the claim, each reported with its mechanism: the corridor harm, the stealth attacker the trust ledger misses, the privileged-teacher ceiling, and the three serving-reliability findings. Within them, the case is made: an audited, shielded, on-device SLM junction controller is possible today, at 0.6B, on a desktop-class 6 GB GPU, with published figures for cabinet-class hardware suggesting but not establishing the same envelope (§7.4).

For a signal engineer the practical reading is narrow. Deploy this where the network can still clear, or where no engineered plan exists at all, rather than on a saturated arterial; calibrate a test network to real demand before believing any delay figure it produces; and treat the audit trail, not the delay, as what certification would rest on. The path onward runs through closed-loop objectives, synthesised shields, and a watchdog-supervised field pilot.

What this study settles is narrower and firmer: fine-tuning does not change that audit trail, which verifies identically in every arm; what it raises is the share of the record the model owns, and it moves delay only where the network can clear, and not on every network that can.

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Chapter A

Reproducibility artifacts

A.1 Environment and version pins

Every measurement in this dissertation was produced under the pins in Table A.1. The Foundry Local version is pinned deliberately: custom-model serving behaviour (registration format, execution-provider options) is version-sensitive, and a mid-campaign upgrade would have invalidated cross-run comparability.

Table A.1: Version and hardware pins.

Component	Pin
Serving runtime	Microsoft Foundry Local 0.8.119
Compile path	onnxruntime-genai model builder, int4 RTN, weight-only
Execution providers	CPU and WebGPU (decision-equivalent, §4.2)
Simulator	SUMO with TraCI, 1,200-step horizon, ≈10 s decision interval
Serving/measurement GPU	NVIDIA RTX 2060, 6 GB (the deployment envelope)
Training GPU (3.8B)	UCL departmental NVIDIA RTX 4090, 24 GB
Training GPU (0.6B)	the RTX 2060 itself (13.5 h)
Host OS	Windows 10 (build 19045), which predates Windows ML EP autoregistration

A.2 Seeds and dataset splits

Training-state seeds: $\{3, 7, 11, 29, 42, 57, 91, 123\}$. Junction states for distillation were sampled only from these. **Closed-loop evaluation seeds:** $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$ gives thirty seeds disjoint from training (§5.3). **Offline holdout:** split by deterministic content hash (`HOLDOUT_MOD=12`) and frozen before the first training run, which leaves 5,580 held-out examples against 59,533 training examples, from 65,113 accepted renderings (6 rejected by the sanity verifier for serving an empty phase without a justifying note). Per format: sota 16,982, myopic 16,982, prediction 18,596, coordination 12,553.

A.3 Prompt contract

All four formats share one output contract: exactly `{"phase": N}`. The *same rendering code serves training, offline evaluation, and closed-loop control*, so no format skew can flatter the offline numbers. The delay-aware (sota) system prompt is reproduced in part:

```
You control ONE traffic-signal junction (this junction only; no neighbour
information). For each green phase you are given: the number of waiting
(queued) vehicles, the TOTAL accumulated waiting time of those vehicles in
seconds, and whether it is the phase currently green. Choose the phase to
serve next to MINIMISE total vehicle waiting time and unnecessary stops --
not simply the longest queue.
```

The myopic format withholds accumulated waiting, prediction adds short-horizon arrival counts, and coordination adds neighbour release notes. Parsing tiers are as given in §3.2.

A.4 The compile-and-serve chain

Reproducing a served custom model requires the following steps in order, each established by failure during this project.

1. Train QLoRA adapters (r=16, NF4 double quantisation, fp16 compute, completion-only loss, family-correct chat template).
2. Merge adapters to a bf16 checkpoint.
3. Compile with the onnxruntime-genai model builder (`-p int4 -e cpu`) using the pinned transformers pairing, because newer environments fail on attention-module or loss-utility imports.
4. Patch `tokenizer_config.json`: `tokenizer_class` must be a value the runtime accepts (`GPT2Tokenizer`), because transformers-5 era configs emit a backend name that fails to load.
5. For Phi checkpoints, strip remote-code `auto_map` entries.
6. **Synthesise** `inference_model.json` (model name plus prompt template). The builder emits none, and without it the model registers but returns HTTP 400 on every request.
7. Copy into the Foundry Local cache directory and, for GPU serving, set `provider_options` to WebGPU in `genai_config.json`.

A.5 Harness invariants

Three properties of the experiment harness underwrite the results. *Fail-loud:* API errors are counted and abort a run rather than being scored as answers. An unservable model records a skip, never a score. *Resumable:* cells are keyed by (model, config, topology, seed) and skipped if already measured, so a crash loses at most one cell. One zero-byte cell caused by a mid-write power loss was detected by JSON validation and re-run. *Recoverable:* after two observed mid-run service self-restarts, the offline evaluator rediscovers the endpoint, reloads the model, and retries failed rows with statistics intact.

Chapter B

Extended results

B.1 Per-topology closed-loop detail

The per-seed cells behind the means and paired tests of Chapter 6 are kept in `results/frontier_raw/`, one file per (model, config, topology, seed) containing the MaxPressure baseline delay, controller delay, relative change, decision composition (SLM / substitution / anti-starvation), from which authored share, acceptance and divergence derive, plus latency distribution and audit-chain check. The fixed-time floor is in `results/experiment_fixedtime_*.json`. Closed-loop figures recompute from both with the scripts in `src/`.

B.2 Failure forensics tables

The catastrophic-cell analysis of §6.7 is tabulated in `results/failure_forensics.md`: 14 of 270 cells at ≥ +20% delay, their seed-matched comparators, and the shared-versus-arm-specific decomposition. The underlying data is in `results/failure_forensics.json`.

B.3 Adversarial trust battery

Per-topology, per-scenario cells for all three controllers (stock and fine-tuned, both scales) are in `results/experiment_trust_traffic_<model>.json`, with raw cells under `results/trust_traffic_raw/` keyed by model, topology, scenario, and seed.

B.4 Authorisation re-test

The four-model, two-arm fair re-test of §6.11 is recorded in `results/slm_auth_lee_<model>.json`, each containing class accuracy, grant accuracy, precision, recall, false-positive and false-negative rates, dangerous false grants, parse failures, and P50/P95/P99 latency per arm.

Chapter C

UK-law fault-grounding knowledge base

The reconstruction layer matches verified collision facts against a curated knowledge base of UK traffic law. The base is deliberately small, explicit, and citable: each rule carries a statute or case reference, a binding strength (`MUST` for statutory prohibitions, `common_law` for judicial doctrine), a fault weight, and the verified fact that triggers it. Table C.1 lists the rules exercised by the seven evaluated scenarios.

Table C.1: Fault-grounding rules exercised by the evaluated scenarios.

Rule	Authority	Effect
Driver crossed red	RTA 1988 s36	Criminal offence; driver fault high
Red prohibition	TSRGD 2016 Sch14	Must not proceed beyond stop line
Amber	TSRGD 2016 5(9)	Driver fault high where a safe stop was possible
EV exemption	TSRGD / conditional	Conditional on an emergency crossing class *plus* either corroboration or a valid emergency-class key; endangerment test still applies
Maintenance vehicle	no exemption	Works vehicle treated as an ordinary driver
Dark signal	Highway Code r.176	Duty to obey falls away; ordinary care governs
Authority misfeasance	*Bird v Pearce*	Positively-wrong indication: authority fault non-zero but low
Authority nonfeasance	*Gorringe v Calderdale*	Mere omission: authority fault defaults to zero
EV apportionment anchor	*Griffin v Mersey*	Default 60/40 against the non-emergency party
Green due regard	TSRGD 2016 Sch14 Pt1 5(14)	Green-light party must proceed with due regard; typically the minority share

One property of the encoding matters for the accountability claim: the emergency exemption is gated first on the mechanically verified crossing class, which a forged key cannot manufacture, and only then on corroboration or a valid emergency-class key, so a spoofed claim leaves the driver squarely within the red-light prohibition (§6.8.1). Legal causation is never assessed but marked `NOT_ASSESSED` for a human adjudicator.